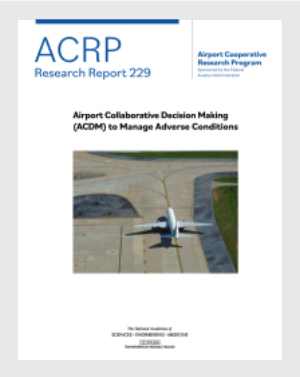
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ISBN 978-0-309-67380-8 | DOI 10.17226/26090

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SUGGESTED CITATION

National Academies of Sciences, Engineering, and Medicine. 2021. *Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions*. Washington, DC: The National Academies Press. https://doi.org/10.17226/26090.

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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 229

Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

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Subscriber Categories
Aviation (AVN) • Operations and Traffic Management (OTM) • Safety and Human Factors (SHF)

Research sponsored by the Federal Aviation Administration

The National Academies of
SCIENCES • ENGINEERING • MEDICINE
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2021

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

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Once selected, each ACRP project is assigned to an expert panel appointed by TRB. Panels include experienced practitioners and research specialists; heavy emphasis is placed on including airport professionals, the intended users of the research products. The panels prepare project statements (requests for proposals), select contractors, and provide technical guidance and counsel throughout the life of the project. The process for developing research problem statements and selecting research agencies has been used by TRB in managing cooperative research programs since 1962. As in other TRB activities, ACRP project panels serve voluntarily without compensation.

Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

ACRP RESEARCH REPORT 229

Project 10-27
ISSN 2572-3731 (Print)
ISSN 2572-374X (Online)
ISBN 978-0-309-67380-8
Library of Congress Control Number 2021931562

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AIRPORT COOPERATIVE RESEARCH PROGRAM

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Transportation Research Board
Business Office
500 Fifth Street, NW
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AUTHOR ACKNOWLEDGMENTS

The research reported herein was performed under ACRP Project 10-27 by a multidisciplinary team led by WSP USA. WSP USA, Inc. was the contractor for this study.

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The authors wish to thank the participants in the survey, interviews, and workshops conducted as part of this project and the stakeholders of airport operations who shared data, perspectives, and materials with the project team. We particularly express appreciation to Sherry Avery (FAA LAX ATCT), Hervé Breton (Groupe ADP), Jack Christine (Charlotte Douglas International Airport), Dave Cunningham (Denver International Airport), Bernie Davis (formerly with American Airlines), Lexie Farmer (Charlotte Douglas International Airport), Sean Fortier (FAA MSP ATCT), Trevis Gardner (McGhee Tyson Airport), Mark Gibbs (Lake Tahoe Airport), Tracy Gillen (FAA DTW ATCT), Bob Hildebidle (FAA MIA ATCT/TRACON), Amber Leathers (Charlotte Douglas International Airport), Eugène Leeman (ACI Europe), Joe Mantello (FAA ORD ATCT), Larry Mares (San Francisco International Airport), Todd Meyer (Division of Aviation of North Carolina DOT), Jeffrey Perry (Charlotte Douglas International Airport), Robert Pyrka (JFK International Air Terminal), Margaret Rendon (FAA SAT ATCT/TRACON), Paul Sichko (Dallas/Fort Worth International Airport), Dave Siewert (FAA JFK ATCT), David Staples (Philadelphia International Airport), Al Stuart (Salt Lake City International Airport), Terri Tibbe (Ted Stevens Anchorage International Airport), Mark Wiebke (Charlotte Douglas International Airport), the Association of California Airports, the North Carolina Airports Association, and the Oregon Airport Management Association.

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FOREWORD

By Theresia H. Schatz
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ACRP Research Report 229: Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions describes processes and tools for airports and their stakeholders for implementation and effective use of CDM to enhance the management of adverse conditions at a variety of types and sizes of airports and aviation systems. It accompanies an electronic toolbox that contains a compendium of implementation practices to apply ACDM principles and tools and identifies the benefits of ACDM as applied to the mitigation of the continuum of disruptions.

This report presents processes for planning and operating with ACDM as well as implementing changes. It features tools to assess the impact on existing practices and provides a comprehensive training program framework. An electronic toolbox was developed that provides information on ACDM and its implementation along with an electronic library on collaborative operations. There are also collaborative tools and templates for information-sharing and decision-making purposes, and reference sheets that document a broad range of adverse events with methods on collaboratively addressing these disruptions.

ACDM is the process whereby airports, airlines, and other stakeholders share information to improve policies, planning, and decisions affecting airport operations. It includes those referenced in *ACRP Report 137: Guidebook for Advancing Collaborative Decision Making (CDM) at Airports* and *ACRP Report 153: Guidebook for IROPS Stakeholder Communication & Coordination*. ACDM is also about developing procedures to anticipate, prevent, and mitigate potential problems arising from adverse internal and external conditions.

CDM has been implemented in the United States for many years, and until very recently this program has mainly focused on cooperation and data sharing between FAA and the airlines. Airports are now at the threshold of becoming fully active partners in CDM. *ACRP Report 137* provides guidance to airport practitioners about the value of CDM and the process for integrating it into operations and planning. This project expands on the research presented in *ACRP Report 137* with a greater focus on the implementation of ACDM and all aspects related to the continuum of disruptions, such as continuity events, irregular operations (IROPS), emergencies, and incidents.

ACRP Project 10-27 was led by WSP USA, Inc. in association with Mosaic ATM, Inc. and Faith Group, LLC, along with the Institute for Transportation Research and Education and Worldwide Aviation Solutions LLC.

The ACDM Toolbox Presentation, the toolbox and its associated documents are available on the TRB website at trb.org by searching on "ACRP Research Report 229."

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CONTENTS

1 Summary

3 Introduction

6 **Chapter 1** Understanding Airport Collaborative Decision Making
6 1.1 Continuum of Disruptions
9 1.2 Introduction to ACDM
13 1.3 Benefits of ACDM
16 1.4 The "C" of CDM
17 1.5 CDM for Everyone

21 **Chapter 2** Operating with ACDM
21 2.1 ACDM Operations Process
24 2.2 ACDM Planning Process
25 2.3 Impact on Current Practices

28 **Chapter 3** Implementing ACDM
28 3.1 Fundamentals of ACDM Implementation
30 3.2 Stakeholder Identification and Executive Buy-In
32 3.3 ACDM Organization
34 3.4 Defining the ACDM Project
37 3.5 ACDM Implementation in Motion

38 **Chapter 4** Featured ACDM Projects
38 4.1 Defining Key Performance Indicators
39 4.2 Daily Coordination Calls and Airport Status Reports
39 4.3 ACDM Website
39 4.4 Integrated Airport Operations Center
40 4.5 Milestone Approach and Departure Metering

41 **Chapter 5** Continuing the ACDM Implementation Effort
41 5.1 Change Management and Promotion of ACDM
43 5.2 Maintaining and Enhancing the Local ACDM

45 **Chapter 6** Certification Framework
45 6.1 Purpose
45 6.2 ACDM Implementation Levels
45 6.3 Certification Process

47 **Chapter 7** ACDM Training Programs
47 7.1 Why Train for ACDM?
47 7.2 ACDM Training Curricula
48 7.3 Operations Modules
49 7.4 Implemention Modules
50 7.5 Knowledge Management

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51 **Appendix A** ACDM Toolbox
53 **Appendix B** Tools and Templates of the Toolbox
54 Bibliography
56 Abbreviations

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SUMMARY

Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

Airport collaborative decision making (ACDM) is a process in which the stakeholders of operations—airport operators, the air traffic control tower staff, flight operators, ground handlers, fixed-base operators, and others—share information to improve policies, planning, real-time coordination, and decisions regarding operations. To achieve this higher level of collaboration, the individuals engaged in ACDM initiatives at individual airports should develop procedures to anticipate, prevent, and mitigate potential operating problems that arise from adverse internal and external conditions. Collaborative decision making (CDM) has been implemented in the United States for many years and until recently has focused mainly on cooperation and data sharing between the Federal Aviation Administration (FAA) and the airlines. Airports are now at the threshold of becoming active partners in CDM. ACRP Project 10-27 investigated how the principles of CDM could apply to airports and developed specific processes, tools, and guidance to implement a scalable and flexible ACDM process at U.S. airports, focusing on aspects related to the continuum of disruptions from continuity events to crisis situations.

Of the airports surveyed as part of this project, 67 percent do not hold regular meetings with the flight operators. Interviews with staff at individual airports show a lack of real-time coordination between the stakeholders. However, nearly all the survey participants responded affirmatively that they would consider holding such meetings to improve collaboration because it is commonly understood that more cooperation can help address local issues and improve overall efficiency. One of the main benefits of operating with ACDM is the mitigation of the "silo effect"—an acute but common symptom of organizational dysfunction that a former Goodyear executive described as a lack of common goals and horizontal communication. Sharing information helps identify weaknesses and deficiencies that might not be detected otherwise. Joint procedures across the stakeholders make the overall operations smoother, increase situational awareness, and enhance both efficiency and resilience.

Research Project ACRP 10-27 developed several tools and methodologies for achieving an advanced state of collaboration between the stakeholders of airport operations. Three operational processes are proposed for enhancing operations planning, real-time operations, and change management. These processes emphasize the need for anticipation, collaboration, and continued improvement at all levels. Along with templates such as the Collaborative Decision Sheets, the operational processes provide a framework for collaboration and a common language for all stakeholders. Operations-proven best practices—such as daily operational briefings—are identified and promoted. These best practices are incorporated in an electronic toolbox, available to practitioners, that features Quick Reference Sheets on various adverse conditions and case studies as well as electronic spreadsheets and dashboards. Together, the toolbox contents cover operational issues that include international flight diversions, the temporary parking of overflow aircraft, best practices on aeronautical information,

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2 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

a must-know on volcanic ash clouds, and lessons learned from recent epidemics and pandemics. Individual airports can customize the toolbox and can upload their own reference sheets, procedures, and plans.

Implementation is a crucial part of any ACDM initiative. A step-by-step approach to achieve ACDM implementation is proposed, supported by templates and a workbook, to involve stakeholders, define common goals and objectives, appoint leadership for the initiative, tailor a vision that serves the local needs, and develop a roadmap of successful projects delivering practical improvements. Techniques are suggested to engage the stakeholders, identify resources, prioritize the projects, gain approval, and run the projects in a collaborative and efficient way. Some of these projects address problems that might be found at various airports, and the most recurrent and relevant ACDM enhancements were inventoried. Beyond the technical and program management aspects, the implementation effort should tackle the cultural change that is essential to ACDM. Training the operations community together on the new processes and tools is an opportunity to do this. A comprehensive joint training program was developed that incorporates several presentation materials available in the toolbox.

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Introduction

Background

The United States has been using the CDM process for many years, and until very recently, this process has focused mainly on cooperation and data sharing between FAA and the flight operators. Airports are now at the threshold of becoming fully active partners in CDM and implementing its principles locally, using the term ACDM. *ACRP Report 137: Guidebook for Advancing Collaborative Decision Making (CDM) at Airports* provides guidance to airport operators about the value of CDM and how to integrate the process into airport operations and planning (Vail et al. 2015).

This project—ACRP Project 10-27, "Using Collaborative Decision Making to Enhance the Management of Adverse Conditions"—aims to develop an electronic toolbox for facilitating the implementation and utilization of ACDM by the stakeholders of airport operations. The electronic toolbox—the first of its kind—provides content about ACDM; guidance on implementation; and tools and templates for fostering collaboration before, during, and after adverse events.

This guidebook presents a selection of narrative contents available in the toolbox to understand the benefits of ACDM for enhancing the management of adverse conditions, to present the main concepts used in ACDM and learn the processes proposed in the toolbox, and to document best practices for implementing ACDM at the airport. A training program is proposed at the end of the report. Training materials for each of the modules are available in the toolbox.

What Is ACDM?

CDM is "a process applied to support activities such as demand/capacity balancing. CDM can be applied across the timeline of activities, from strategic planning to real-time operations. CDM is not an objective but a way to reach the performance objectives of the processes it supports" (ICAO Doc 9971, 2018).

Behind these high-level objectives, CDM typically consists of two key ideas:

- Air traffic management (ATM) should be based on the same flight data updated in real time and shared between the different stakeholders. However, information sharing alone is not sufficient to realize CDM and the objectives of CDM.
- Decision-making should be coordinated and collaborative, especially during adverse conditions, including crisis. Collaborative operating procedures should be defined to facilitate the decision-making process and crisis management.

ACDM is the application of this approach to airport operations. This guidebook and the toolbox focus on the second part of CDM for airports: the enhancement of the decision-making process for managing adverse conditions.

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4 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

Airport collaborative decision making is abbreviated "A-CDM" in most parts of the world. It is the acronym used by the International Civil Aviation Organization (ICAO) and industry organizations when talking about the concept of airport CDM.

The similar acronym ACDM was used in *ACRP Report 137* (Vail et al. 2015). The current report uses this acronym when referencing airport CDM in the United States in order to differentiate it from other ACDM initiatives that might be more ATM-centric (e.g., EUROCONTROL).

Why Are the Guidebook and the Toolbox Needed?

While CDM emerged more than two decades ago in the United States, the process has focused on flight operations from takeoff to landing. Collaboration in airport operations is still an ongoing effort as well. It has progressed at some of the Part 139 certified airports with the implementation of collaborative frameworks for irregular operations management and contingency planning based on guidance provided by *ACRP Reports 65* (Nash et al. 2012) and *153* (Cogliandro et al. 2016). However, continuity events are not covered by these processes, and interviews with airport operators and stakeholders confirmed the lack of direct coordination during regular operations.

Nearly all participants in the survey and interviews conducted during the project called for more cooperation in airport operations management. This guidebook and toolbox promote collaboration at all levels for increasing operational resilience. They are a timely response to a strong need for tighter cooperation in operations planning and real-time operations.

What Is the Purpose of the Toolbox?

The ACDM toolbox serves as a go-to manual for airport operators and their stakeholders for effective implementation and use of CDM to enhance the management of the continuum of disruptions at a variety of types and sizes of airports.

The toolbox addresses and includes the following:

- A compendium of implementation practices to apply CDM principles and tools as they apply to the continuum of disruptions,
- Identification of the benefits of CDM as applied to the mitigation of the continuum of disruptions,
- The use of data sharing to drive internal and external communications,
- Techniques for engaging stakeholders with an emphasis on airline partners at the system operations level,
- An effective approach to standardization,
- A consolidated glossary of key and new common terminology,
- Guidance on how to use CDM to ensure scalability and flexibility across all types and sizes of airports for managing the continuum of disruptions, and
- Templates for communication criteria for international operational diversions.

The toolbox also features a comprehensive joint training program for the local ACDM sponsors, project managers, and practitioners as well as several resources on collaboration in airport operations.

Who Should Use the Guidebook and the Toolbox?

The guidebook and toolbox are intended to be used by all the stakeholders of airport operations who are looking for (1) information on ACDM and its benefits, (2) guidance to implement ACDM locally, (3) guidance to more generally enhance collaboration between stakeholders, and (4) resources on adverse conditions and how to manage them collaboratively. Typical users

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are the internal stakeholders of airport operators—including airport operations, emergency and irregular operations management, and other divisions—and the partners of real-time operations, which include air traffic controllers, flight operators, ground handlers, fixed-base operators, tenants, and federal and law enforcement agencies.

While the ACDM benefits might seem more evident for hub airports of the National Airspace System, the contents featured in this guidebook and toolbox can be used to implement ACDM and improve resilience at all aviation facilities (including smaller commercial service airports and general aviation facilities) and to foster collaboration within multi-airport systems, regional groups of airports with interdependencies, and state aviation systems.

Moreover, the materials are not intended to be only used for implementing a formal ACDM organization. The guidance and best practices provided can help aviation operations stakeholders to simply enhance collaboration without entering into any kind of formal agreement and developing an ACDM initiative. Even so, when feasible, implementing ACDM practices is recommended to maximize the benefits in the long term. The toolbox should be used as a resource to develop collaboration "à la carte" to address local issues on operations efficiency and resilience.

How to Use the Guidebook and the Toolbox

The guidebook features a selection of narrative contents from the toolbox. They provide information and case studies on ACDM and, more broadly, on collaboration in airport operations for enhancing the management of adverse conditions. They propose collaborative processes to develop plans for adverse conditions, implement these changes successfully, and operate with ACDM in real time. They dispense best practices for ACDM implementation and project management and a step-by-step approach to conducting the ACDM initiative. A comprehensive training program is available to educate the operations community about ACDM and train each population of the ACDM organization, depending on its role, with a tailored set of modules.

The toolbox application is the main deliverable of this ACRP Project 10-27. Through the main functions—Learn, Implement, Train, and Operate—users can progressively increase their knowledge about ACDM and have the knowledge and tools they need to contribute to the local ACDM initiative depending on their role within the ACDM organization. This toolbox also has a set of Quick Reference Sheets, tools, and templates listed by categories that can assist the operations community in addressing operational issues in real time. Finally, it also contains a glossary, abbreviations, and an electronic library that cover operations matters beyond ACDM. Finally, this guidebook encourages individual airports to partially customize the toolbox in order to disseminate the collaborative procedures and plans and other useful information about their operations to their members. Consequently, this toolbox is intended to be used as a practice-ready, straightforward "Swiss Army knife" by the airport operations community.

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CHAPTER 1

Understanding Airport Collaborative Decision Making

1.1 Continuum of Disruptions

What Is the Continuum of Disruptions?

The daily life of an airport is never like a long, quiet river. The path of regular operations is always full of events of various types and amplitudes that will cause disruptions. Moreover, one disruption can lead to another, and another disruption can happen while an ongoing disruption is still in the process of being addressed. To be resilient, airport managers and their stakeholders should think of operations as a complex continuum of disruptions rather than as a peaceful continuum of operations (Figure 1).

While some adverse events cannot be predicted before they occur (e.g., runway overrun), many of them are preceded by precursors that, if detected, can help the community of airport operations to take proactive steps that will either expedite the response or minimize the impact of the event (Figure 2). For instance, snowstorms can usually be predicted by weather forecasters. Pre-positioning equipment and mobilizing drivers in anticipation of the snowfall makes these means available for pre-treatment and immediately when the snow starts falling.

Characterizing the Criticality

However, each disruption of regular operations is different in the way it affects the airport. Also, the disruptions do not all call for the same type of response and mobilization of resources. Despite these dissimilarities, the disruptions can be categorized based on their criticality, and in particular, by (1) their impact on operations and (2) the resources needed to address them (Table 1).

The following four levels of disruptions are used in the toolbox and its materials:

- A **Continuity Event** is a planned or unplanned loss of a system that impedes operations but does not significantly affect most flights (e.g., partial loss of airfield lighting, limited IT system outage, and limited staffing shortage). Continuity events can be addressed by the airport staff as part of their daily missions and typically require some coordination with a small number of stakeholders who follow standard operating procedures.
- **Irregular Operations (IROPS)** are exceptional events that require actions and/or capabilities beyond those considered usual by aviation service providers but that are still manageable by airport-based resources. Generally speaking, the impact of an IROPS event is manifested when passengers experience delays, often in unexpected locations, for an undetermined amount of time. IROPS include ground-delay programs and ground stops, massive diversions, tarmac delays, snowstorms, and taxiway excursions. IROPS are typically covered by airport IROPS contingency plans, which establish the coordinated response of the airport and its stakeholders.

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Figure 1. Depiction of the continuum of disruptions.

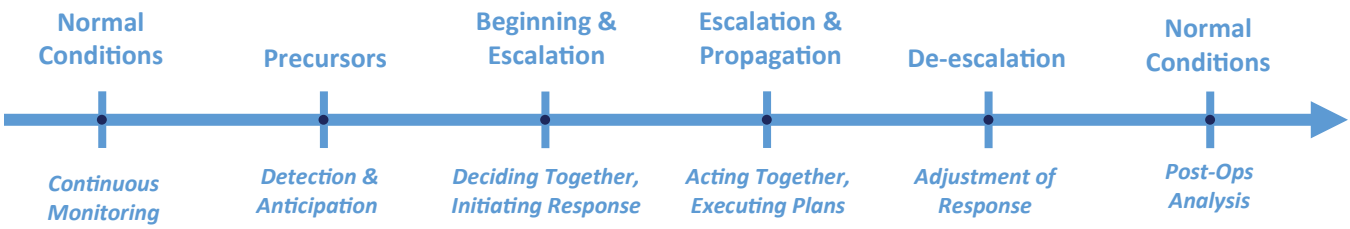


Figure 2. Timeline of adverse conditions and the typical response.

Table 1. Criticality of disruptions.

Criticality	Typical Impact	Resources Involved	Examples
Continuity Event	Limited impact on some airport functions and/or flights	Airport-based resources are enough	▪ Individual flight delay ▪ Flight Information Display System screen out of service ▪ Bathroom unavailable
Irregular Operations	Flight delays/missed flights		▪ Massive diversion ▪ Ground delay program/ground stop
Emergency	Dispatch of first responders		▪ Conflict situation ▪ Medical emergency ▪ Intrusion in a security-restricted area ▪ Suspicious bag
Incident/Crisis	Systemic disruption significantly impacting most of the operations	Additional non-airport-based resources required	▪ Aircraft accident ▪ Hazardous material spill ▪ Terrorist attack ▪ Wildland or urban fire ▪ Flooding ▪ Earthquake ▪ Hurricane ▪ Tornado ▪ Tropical storm ▪ Pandemic ▪ Power outage

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8 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

Other plans and procedures, such as a Snow and Ice Control Plan, provide additional planning and procedures on specific matters.

- An **Emergency**, in this context, is any occasion that warrants action to save lives and to protect property and public health with the dispatch of first responders (e.g., paramedics, emergency medical technicians, and airport police officers) as long as they can be managed by the pre-determined (organic) response assets without exceeding airport capabilities. This includes but is not limited to conflict situations, medical emergencies, intrusions in a security restricted area, and suspicious bags.
- An **Incident or Crisis** is an occurrence or event (natural or human-caused) that requires an additional, non-airport coordinated emergency response to protect life, property, environment, or business operation. For example, incidents can include aircraft accidents, major disasters, large-scale emergencies, terrorist attacks, terrorist threats, floods, hazardous materials spills, earthquakes, hurricanes, tornadoes, tropical storms, public health emergencies, and other occurrences that require an emergency response or an enhanced response by supporting subject matter experts beyond the airport's resources. The main airport emergency planning document for incidents and crises is the Airport Emergency Plan, which addresses essential emergency-related and deliberate actions planned to ensure the safety of and emergency services for the airport populace and the community in which the airport is located.

Categorization of Adverse Conditions

Whatever the level of disruption, it is essential for the airport community to anticipate any event and to plan accordingly as much as possible. While not all events can be predicted or require a comprehensive response plan, all of them can be categorized. Criticality regarding the impact on operations and the resources needed to provide an adequate response has already been discussed. Another way to characterize events is to refer to their area of impact (e.g., airside/airspace, terminal, landside, and off-airport) and their domain (a type of event or main function affected). The toolbox adopts the following categorizations (Figure 3). Other categorizations might be possible, and some events might fall under more than one area of impact/domain pair. However, the proposed system aims to provide a logical and intuitive way of facilitating information searches.

Planning and Deciding Together

To mitigate the consequences of adverse events, the airport should deliver an adequate response in order to return to the path of regular operations as soon as possible. This process requires

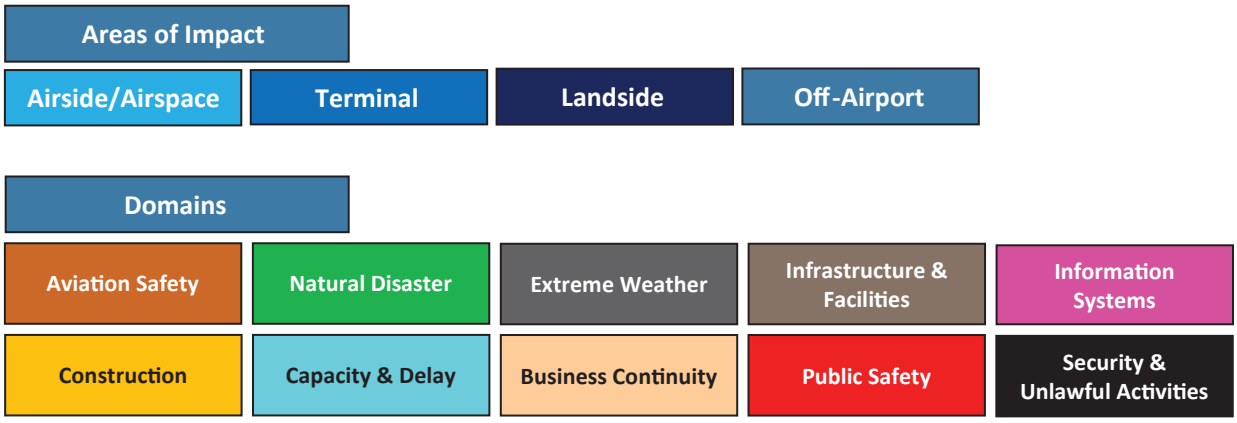


Figure 3. Events and items categorization adopted in the toolbox.

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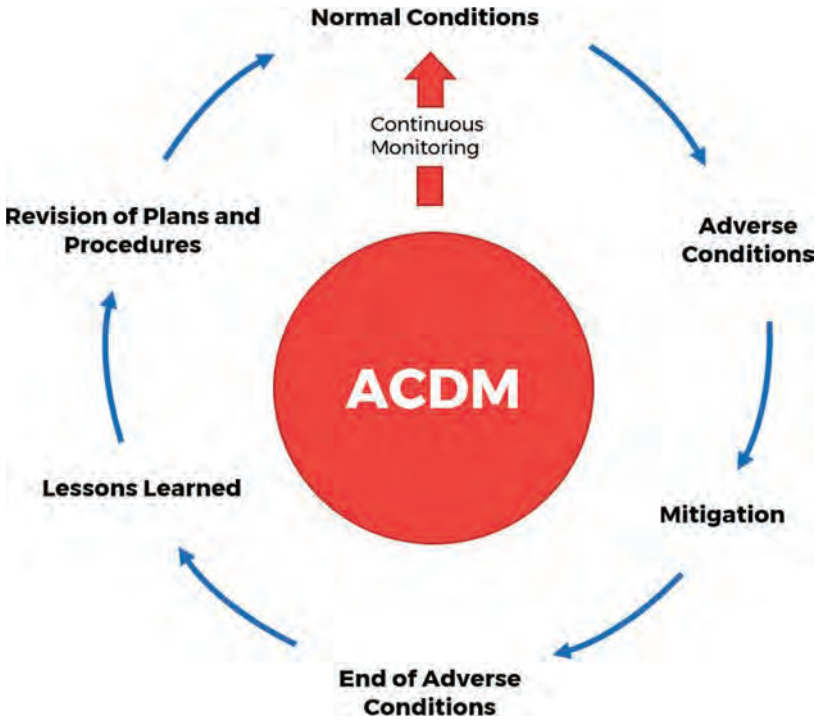


Figure 4. Life cycle of the management of adverse conditions under ACDM.

correctly identifying potential disruptions in order to plan the response and establish procedures with the concerned internal and external stakeholders, to arm the response measures, and to oversee them. Collaboration between the stakeholders who have different profiles and cultures is the key to establishing successful plans and procedures that make sense operationally and can effectively contain and address adverse conditions.

Airport Collaborative Decision Making (ACDM) is a process of excellence that aims to enhance airport operations to manage non-nominal situations (Figure 4). Each stakeholder optimizes decisions based on shared, reliable, and continuously updated information, especially in cases of predicted or unpredicted events. With this process, the airport can prevent any degraded situation and mitigate efficiently by continuously monitoring performance indicators or disruption premises. It is important to alert the respective concerned persons to avoid the accumulation of small events that can cause an emergency or even a crisis.

The time from the end of the disruption to the return to normal conditions is a fragile and degraded period that can lead to incidents. Analyzing feedback after the event needs to be done properly to enhance management procedures or even to create them to face a future crisis. It has to be done internally, as well as with all the stakeholders in workgroups, as part of the CDM process to present the results and decide on the major updates to existing procedures.

1.2 Introduction to ACDM

Definition

CDM is "a process applied to support activities such as demand/capacity balancing. CDM can be applied across the timeline of activities, from strategic planning to real-time operations. CDM is not an objective but a way to reach the performance objectives of the processes it supports" (ICAO Doc 9971, 2018).

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Behind these high-level objectives, CDM typically consists of two key ideas:

- Air traffic management (ATM) should be based on the same flight data updated in real time and shared between the different stakeholders. However, sharing information by itself is not sufficient to realize CDM and its objectives.
- Decision making should be coordinated and collaborative, especially during adverse conditions, including crisis. Collaborative operating procedures should be defined to facilitate the decision-making process and crisis management.

The present toolbox specifically focuses on the second key idea and CDM per se for enhancing the management of adverse conditions. More information on the benefits of CDM for real time ATM is available in *ACRP Report 137: Guidebook for Advancing Collaborative Decision Making (CDM) at Airports* (Vail et al. 2015).

A Brief History of CDM in the United States

In the mid-1990s, as delays in the National Airspace System (NAS) grew, the Federal Aviation Administration (FAA) and flight operators came to realize that internal real-time operational decisions had a great deal of external impact on one another. As this understanding deepened, they realized that decisions were being made without full and complete information and collaboration. To bridge this gap, the FAA/Industry CDM Stakeholders Group was established to collaborate and share real-time operational information to improve situational awareness and decision making.

This ATM collaboration has now evolved to include airport surface traffic management. This FAA/Industry CDM activity has realized that comprehensive industry collaboration must extend to the aircraft parking gate. Thus, airport operators are also a vital link in the effort to reduce delays and improve safety and efficiency (Figure 5).



ACDM is coming: Several U.S. airports have effectively developed practices and tools for sharing real-time information and collaborative decision-making purposes. For instance, Dallas/Fort Worth International Airport has implemented a collaborative approach to massive diversion with the stakeholders and the surrounding airports. JFK International Airport is consolidating its operations based on the 2018 LaHood report that recommends more integration and collaboration.

The Case for ACDM

ACDM is the extension of the CDM philosophy of information sharing to problems affecting airports. In an ACDM context, there are generally three distinct leadership groups: the airport

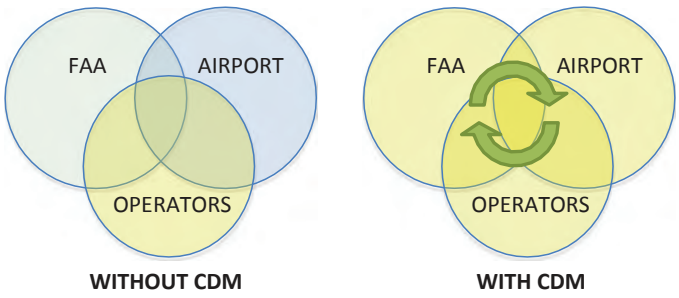


Figure 5. Airport functions with and without CDM.

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operator, the FAA (Air Traffic Control), and the flight operators—a category that includes airlines, air taxis, and private pilots. Other stakeholders, such as the ground-handling service providers and law enforcement agencies, may also play a critical role in the ACDM process. Each of these groups represents different constituents as they strive for efficient, timely operation and work to solve issues related to aircraft surface operations at airports.

Before the implementation of ACDM locally, these leadership groups at the airport worked in "silos." Exchange of real-time, day-of-operations information among the groups was limited. With ACDM, all stakeholders optimize their operations and decisions in a collaborative environment, with consideration of their preferences, known constraints, and forecast situations. The decision-making process is facilitated not only by sharing accurate and timely operational information through a common toolset, but also by applying agreed-upon process and procedures (Table 2).

For the management of adverse conditions, ACDM relies on the following:

- Using shared, practice-ready procedures to guide the action of the stakeholders and make it more efficient.
- Addressing operational disruptions together, not in "silos."
- Anticipating, preventing, and preparing for adverse conditions as far as practicable.
- Mitigating adverse conditions early on, before they degenerate into a more impactful crisis.
- Returning to regular operations as soon as possible.

Table 2. Collaborative approach and results examples.

Collaborative Approach	Collaborative Result
Strong coordination among the stakeholders during winter operations (e.g., joint training/briefings before each winter, integration of airport operations, airline representative at the Airport Operations Center, shared tools/data with the air traffic control tower, and runway closures for snow removal tightly coordinated/planned together).	▪ More efficient snow removal/pavement de-icing operations, ▪ Increased safety for both winter operations crew and the aircraft, ▪ Enhanced passenger experience, and ▪ Overall reduction of delays due to winter conditions.
Collaborative procedure with the stakeholders on disabled aircraft removal including pre-identification of removal equipment and plan for mitigating impact of runway unavailability on operations.	▪ Higher awareness of stakeholders about roles and responsibilities, ▪ Shorter decision-making process and faster removal of aircraft, and ▪ Reduced delay/impact on operations.
Collaborative approach to runway construction projects. Construction Safety and Phasing Plan and Safety Risk Assessment involve the stakeholders and the construction management leadership. Regular (weekly and daily) construction coordination meetings in place. "Plan B" prepared for the beginning and end of the most impactful construction projects.	▪ Less delay due to modified ground routing or runway capacity, ▪ Increased awareness of the pilot community and air traffic controllers, ▪ Safer conditions on the operative airside/airspace, and ▪ Increased operational robustness.
Collaborative procedure with neighboring airports on the diversion of international flights.	▪ International flights are not diverted to airports without Customs and Border Protection unless agents can be positioned there on short notice, ▪ Reduced waiting time of passengers inside the aircraft, ▪ Enhanced passenger experience, and ▪ Easier transfer of passengers to their final destination.

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Toward the ACDM Process

While many airports have some kind of collaborative practices and procedures in place with their stakeholders, few of them have a comprehensive approach that can be considered an established ACDM process. The opportunity with ACDM is to bring all the aspects of airport operations under an integrated process of decision making that gathers both the internal and external stakeholders of operations. This integrated vision of operations can help in creating synergies, identifying issues that are not adequately addressed, and performing operations planning with all the concerned parties for making the airport more proactive and resilient. It also implies the establishment among the stakeholders of a common definition of airport performance, joint efficiency and resilience objectives, and a framework for continuous improvement.

The ACDM process is intended to encompass the management of all regular and irregular operations with a specific focus on capacity and efficiency. Consequently, ACDM does not supersede other processes with collaborative features, such as the airport safety management system or the airport framework for emergency management, but provides a different focus. However, it might become an umbrella term that embraces and coordinates existing processes with a similar focus, such as the IROPs Contingency Plan.

ACDM in a Nutshell

ACDM is a process, not a project. To recap, it requires the following elements:

- First and foremost, everyone needs to realize that a better airport operation benefits everyone. Many issues should be addressed by the whole community of operations together rather than individual entities side by side in silos.
- The ACDM process at a specific airport should first address local needs and serve the vision of the local community of operations, and then help the airport to enhance its performance within the NAS.
- Success will require the commitment of airport resources from the operating staff to the upper management. ACDM is not a quick fix. It is a process that requires time and commitment. However, it can significantly enhance operations and make the whole airport more resilient during adverse conditions.
- All affected internal and external stakeholders need to be included, even those who have little technical knowledge of the issues being addressed. Their inclusion is essential to ensure the ACDM process is comprehensive and efficient in addressing a broad range of operational issues.
- Team building and trust must be established and strengthened. Once it becomes a reality, it is crucial to maintain this "ACDM spirit" through recurrent training, seasonal workshops, and regular meetings. Joint procedures, data sharing, and group goals are for the benefit of the entire airport, not single entities.
- Data availability must be safeguarded and protected from use for competitive reasons between participating entities. Useful data are available in every airport operation, and transitioning the data to usable information is a group effort.
- Sharing of knowledge and experience is critical. Airport and stakeholder staff have valuable experience and knowledge, and different perspectives can ensure a valuable end product that benefits everyone. This includes sharing of ACDM experience and lessons learned between airport staff as well as among stakeholders at industry events.
- Consensual metrics should be defined and utilized to measure success. Even though a formal cost-benefit analysis will not usually be required, success and benefits need to be measured to improve the next application and pave the way for the future development of the local ACDM process.

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- Decisions made and data gathered during adverse conditions should be recorded; the toolbox proposes different recording tools. Regular meetings should be held for debriefing these events, identifying lessons learned, and enhancing the ACDM process—especially individual procedures used during such events. Lessons learned should be tabulated for future efforts and shared within the community of operations.

1.3 Benefits of ACDM

Advantages and Efficiencies for Managing Adverse Conditions

Many potential advantages and efficiencies can be realized through ACDM-enabled projects. A few of these are listed below:

- **Early warning and anticipation.** Under ACDM, the stakeholders agree on monitoring specific key performance indicators and share information on potential disruptions as soon as possible. They inform each other through daily operations calls. They proactively manage adverse conditions and coordinate the response when applicable.
- **Operations resilience.** Under ACDM, the stakeholders have predefined procedures covering a broad range of potential operational issues and mutual knowledge of their respective organization and constraints, and they coordinate their response and find synergies as far as practicable.
- **Predictability and resource management.** At commercial service airports, sharing real-time flight operations data facilitates the predictability of the passenger and aircraft flows and consequent decisions on resource allocation and capacity adjustments.

Indirect benefits include the following:

- **Brand enhancement.** A more efficient airport may generate more passengers and additional flights, both of which generate airport revenue. Brand enhancement was identified as important in several airport interviews conducted under ACRP Project 10-19 [*ACRP Report 137* (Vail et al. 2015)].
- **Real-time diversion notification.** If airports are afforded access to FAA/Industry CDM information, real-time notice of airline diversions can be gained through the Aggregate Demand List or Diversion Recovery CDM pages. Thus, airports have immediate notification of actual flight diversions and more time to plan for these operations.
- **Improved operational data.** Data are available from several sources to completely analyze the entire surface movement history of a flight for post-event analysis, thus identifying possible changes to increase efficiency and safety.
- **Scheduling and service improvements.** With real-time estimated arrival times, airports could forecast well in advance the demand for passenger services, Customs and Border Protection, baggage, food services, and other services and activities. The status and number of late flights could be relayed to airport concessionaires.
- **Construction mitigation planning.** History has shown that the planned mitigation for a number of construction projects has been altered and improved by collaborative inclusion of the operator preferences.

At relatively uncongested airports, one of the prime considerations is departure readiness prediction to enable more efficient operations to merge into major traffic flows or streams. For example, smaller airports generated many short-range flights into congested airports. The congested airport arrival traffic is time-based metered by FAA. Time-based metering results in each flight being assigned a metered time to provide a smooth, constant flow of arrival traffic. Short-range flights are the last to call for metering times, and thus metering time availability is limited, which results in the departure being held on the origin taxiway for its metering time.

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If accurate flight departure readiness predictions could be made 30–60 minutes in advance, metering time requests could be made earlier, which could result in delay absorption at the gate before passengers board the aircraft, and in some cases, could reduce the amount of delay required.



What are the benefits of ACDM for addressing adverse conditions?

- Uses shared procedures to guide the actions of the stakeholders, making actions more efficient.
- Addresses operational disruptions together, not in silos.
- Anticipates, prevents, and prepares for adverse conditions as far as practicable.
- Mitigates adverse conditions early, before they degenerate into a more impactful crisis.
- Returns the airport to regular operations as soon as possible.

Better Management of Airport Resources

Coordination and information exchange can provide airports with more accurate information on resource use. Better information on short-term gate needs can help with tactically allocating gates to meet demand, while historical data on gate and taxiway usage can help both to plan long term and to monitor conformance with airport use agreements. Sharing information on airport construction helps to mitigate its impact on capacity and operations.

In adverse conditions, data sharing may alert airport operators and their stakeholders to events that put a strain on their resources, such as multiple diversions heading to the airport in a short period. With more advanced awareness of the situation, airport operators can more effectively plan their response and utilize their resources in a way that minimizes the disruption to regular operations and to the passengers. With greater coordination among stakeholders, airports can recover more quickly from events that disrupt normal operations (such as snow or fog). This reduces the delays experienced by passengers and increases the throughput of the airport.

Further ACDM Benefits

The specific benefits will greatly vary from one airport to another, depending on the ambitions of the community of operations and the objectives of the local ACDM process. Several of these benefits would require participation by the air traffic control (ATC). At larger airports, this may preclude some of the benefits from being realized, or at least fully realized, through commercial ACDM applications intended primarily for use by the airport staff. In general, the greater and more ardent the formal participation of the airports' stakeholders, including ATC, the broader the range of potential benefits.

Other ACDM applications may also help to enhance future airport planning and development efforts, which usually result in construction activity. For instance, ACDM-type activities have helped mitigate construction impacts in New York City, Atlanta, Memphis, and other locations. In New York City, departures were metered to control runway queues; and in Atlanta and Memphis, construction vehicle traffic across active taxiways or ramps was controlled and metered to ensure safety around taxiing aircraft.

Any ACDM effort will inevitably involve data sharing. If the appropriate data are shared in such a way as to allow straightforward means of access in specific locations (e.g., an airport-operated ramp

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tower), operational performance should improve. Using the San Francisco International Airport major runway construction in 2014 as an example, FAA, in collaboration with many stakeholders, provided an ACDM-type data-sharing feed via the System Wide Information Management system to aid in situational awareness and to control departure queues. As with the FAA/Industry CDM program, FAA did not provide the software to display the data; it only provided the data.

Additionally, ACDM data sharing results in post-operational analysis. Having a better understanding and record of both typical and unusual operational trends will enable better planning. Better planning will result in improved operational predictability.

Finally, several airports and airlines operate or contract out ramp control towers. These facilities were originally established to better control the vehicle and aircraft movement activities in the non-movement area. But their scope and responsibility have expanded to include, in some cases, major portions of the airport surface, including staging departures in a more efficient sequence (i.e., United Airlines Ramp Tower at Newark Liberty Airport). The gate-to-gate concept long advocated by EUROCONTROL and the similar concept envisioned by the FAA Terminal Flight Data Manager (TFDM) program will certainly utilize both ramp towers and operational control centers.

Role of the Stakeholders of Airport Operations

ACDM is a process to identify problems and build solutions through consensus planning, information sharing, and application of any needed information technology tools. The airport operator plays a critical role in developing and implementing the local ACDM implementation roadmap. The airport operator may be the ACDM initiator and lead the effort, may have a lead partnership role with a major user of the airport, or may be the facilitator or collaborator as part of the effort.

For instance, at Hartsfield-Jackson Atlanta International Airport, Memphis International Airport, and Dallas/Fort Worth International Airport, the airport operator has partnered with the major air carrier and/or research activity for CDM-like operations. The Port Authority of New York and New Jersey assumed the lead role in the departure metering project at New York City's John F. Kennedy International Airport (JFK). San Francisco International Airport (SFO) requested and received the Airport Surface Surveillance Capability (ASSC) viewer data from the FAA to help mitigate the impact of a runway closure.

Each stakeholder's role may also be determined by the density and complexity of operations at the airport. While many of the operational issues, such as safety and mitigating environmental impacts, are the same, there are some distinct differences. For example, at commercial service airports with several airlines, the local Airlines Operations Committee or a designated committee member airline might represent a coalition of minor air carriers in the ACDM process. The main air carrier(s) might be directly participating in the process.



Who supports ACDM?

ACDM is supported by virtually all the stakeholders of the airport industry worldwide: Airports Council International, Civil Air Navigation Services Organization, International Air Transport Association, International Federation of Air Line Pilots' Associations, and International Civil Aviation Organization. They have recognized the benefits of ACDM to aviation, and they have published recommendations for and positions in favor of implementing CDM.

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1.4 The "C" of CDM

Collaboration as a Core Value

The most significant and powerful aspect of CDM is the in-depth collaboration it implies and promotes. Implementing ACDM is, first and foremost, a cultural change. It is about agreeing on and embracing the CDM approach, building mutual trust between the stakeholders, sharing operational information transparently, and making decisions together for enhancing operational efficiency and robustness.

Airport stakeholders have always worked to maintain a good relationship and coordinate on major issues. However, this coordination is not necessarily based on permanent communication channels and set procedures, making it harder for the stakeholders to identify, anticipate, and mitigate together the impact of adverse conditions. ACDM offers a formal framework to share information transparently in real time, to agree on a strategy, to make decisions together on operational issues, and to continue improvement through a lessons-learned process.



Airports want more collaboration: Of the airports surveyed as part of the ACRP 10-27 project, 67% do not hold regular meetings with the flight operators. Interviews with individual airports show a lack of real-time coordination among the stakeholders. However, nearly all the participants to the survey responded affirmatively that they would consider holding such meetings to improve collaboration. There is a common understanding that more cooperation can help address local issues and improve the overall efficiency.

The "Silo Effect"

The stakeholders of airport operations understand that operational challenges are highly demanding. Under increasing traffic, heightened security standards, and emerging technologies, simply maintaining day-to-day activities is becoming increasingly difficult. Resilience is at risk, and any adverse conditions might cause unacceptable disruptions in the operations. To prevent and mitigate such operational disruptions, it is essential that airport stakeholders collaborate to share information, raise common awareness, and determine together the best approach to prevent or mitigate disruptions by sharing resources if necessary.

However, they often operate independent processes and isolated systems, focusing on their own outcomes and without a shared situational awareness across the wider airport community. In other words, they work side by side (the silo effect) instead of together. This limited perspective on the operation as a whole can result in widespread inefficiencies and even conflicting and opposite effort.



Phil S. Ensor, a former director of Goodyear Tire and Rubber Company, explained in ***The Functional Silo Syndrome*** the organizational difficulties of employees working without cross-communication: "[P]eople across the organization do not share common goals. Their goals are primarily functional. Communication is heavily top-down—on the vertical axis. Little is shared on the horizontal axis, partly because each function develops its own special language and set of buzzwords" (Ensor 1988).

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The roots of the silo effect at airports are typically:

- **Working in isolation.** Each stakeholder runs processes that serve the stakeholder-specific priorities and objectives. Parties concerned with the same issue do not necessarily discuss their concerns together, and they try to address it with different strategies that can be conflicting and overlapping.
- **Lack of common vocabulary and definitions.** Each stakeholder uses terminology that is not fully understood by others, preventing transparency and impeding the definition of common objectives. For instance, one airline's definition of "departure on time" might be different from that of other air carriers, the air traffic control tower, and the airport operator.
- **Lack of information exchange and communication.** Stakeholders do not exchange information on flight operations, airport and external conditions, and airport resources. For instance, the airport operator is not adequately informed about flight diversions and is not aware of them before they land at the airport.
- **Lack of joint strategic vision.** No common understanding and joint ambitions exist among the stakeholders of airport operations.

The CDM Philosophy

In an ACDM environment, internal and external stakeholders of airport operations share information on flight operations, airport and external conditions, and airport resource availability in real time. They hold daily meetings or calls to discuss the coming hours of operations and tactical coordination. They decide together about any matters that affect capacity, resiliency, and safety, which are considered common, jointly managed resources. They organize regular (e.g., quarterly) ACDM meetings for analyzing performance, sharing lessons learned, and deciding about further improvements.

The CDM philosophy demands the following from every stakeholder:

- **Commitment to the ACDM process.** The stakeholders should agree on the local benefits of ACDM, define together the objectives of the local ACDM initiative, and embrace this project.
- **Information sharing and transparency.** The local ACDM project should involve all the stakeholders at the same level of decision—symbolized by the Borromean rings of the CDM@USA visual identity in the toolbox. Policies and procedures should be available to all the stakeholders. Information should be shared in real time among all the stakeholders.
- **Deciding together.** Decisions affecting more than one stakeholder should be made together based on collaborative procedures and communication channels. The overall airport capacity, resilience, and safety should be considered common resources.
- **Lessons learned and continuing improvement.** The ACDM process should build on previous adverse events and common experience. Decisions and actions should be debriefed. Lessons learned should be shared, and improvements to operations management should be discussed as needed.

1.5 CDM for Everyone

Scalability and Applicability of ACDM

Because the ACDM process should address local needs first and serve the vision of the local community of operations, ACDM efforts will be different at different airports. The ACDM process can be scaled to large, medium, or small hubs, joint-use airports, and general facilities in the U.S. European efforts have proven this scalability of ACDM with successful applications at airports of various size from large hubs (e.g., Paris-Charles de Gaulle Airport, Frankfurt Airport,

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Heathrow Airport, Madrid-Barajas Adolfo Suárez Airport, and Zurich Airport) to mid-sized airports (e.g., Lyon-Saint Exupéry Airport, Milan Linate Airport, Napoli Airport, Palma de Mallorca Airport, and Venice Airport Marco Polo), and a regional CDM is now under development for integrating smaller airports. They also show that ACDM can become a system-wide standard and improve operations efficiency and resilience at the scale of a large aviation system similar to the U.S. NAS.

While there is a common path to ACDM, there is no single answer to what the ACDM process and organization are supposed to be intrinsically, because they have to serve the specific needs and vision of an airport and its community of operations. As the proverbial saying goes, in this context: "when you've seen one airport, you've seen one airport." This is true for ACDM as well. However, it is possible to build a standard approach to the core objectives of ACDM, the current framework for airport operations and emergency management, and the lessons learned from the existing ACDM process or ACDM-like practices. Chapter 3, Implementing ACDM, details this standard approach.

From Curbside to Airspace

Several hub airports have or are in the process of procuring advanced IT solutions for airport operations management that include the exchange and treatment of real-time flight data with the stakeholders. It is likely that deploying ACDM at such airports will have an ATM component, as outlined in *ACRP Report 137* (Vail et al. 2015). However, ACDM is not synonymous with airside and airspace only. The ACDM approach and its benefits can apply to all airport components, from the curbside to the terminal control area and beyond with the FAA/Industry CDM. This integrated vision of collaboration is called Total Airport Management.

A powerful vehicle of the Total Airport Management approach at larger commercial service airports is the integration of the real-time operations management functions into an integrated Airport Operations Center with a "one-roof" vision. Such centers are sometimes called Airport Operations Centers, Integrated Airport Command Centers, Integrated Airport Operations Centers, Integrated Airport Management Centers, or Integrated Operations Centers.

CDM at Hub Airports

CDM initiatives at hub airports typically involve a large number of stakeholders, including the different airport departments, the FAA Air Traffic Control Tower (ATCT), the ramp control tower(s), the air carriers and any local hub control centers, and the ground handlers. Depending on the ambitions of the local community of operations and the defined objectives for the ACDM implementation roadmap, it can bring together several individual projects for enhancing airport operations. For instance, over 100 projects were included in the CDM@CDG 2020+ Roadmap at Paris-CDG, France. The implementation of ACDM at hub airports might require creating a specific project management structure in charge of running this implementation as a project and then transferring it to the operations when complete.

When the first elements of ACDM are implemented and the process starts to be operated, it is important to keep the "ACDM spirit" alive and have a process of continuous improvement based on lessons learned and new needs. ACDM is also the perfect platform to hold performance and efficiency meetings with the stakeholders for reviewing key performance indicators and address any hindrances. For this purpose, CDM Champions should be designated within each stakeholder organization for championing ACDM internally (e.g., disseminating important information and reporting to the ACDM Manager). Different workgroups and committees might be needed as well. They include at least an ACDM Steering Committee to determine the

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operational objectives of the ACDM process and the means allocated to its operation and an ACDM Operations Committee to work on streamlining the process. These groups are detailed in Chapter 3, Implementing ACDM.

Small is beautiful. "Smart airport" is not a designation limited to large-hub airports. "Lite" IT solutions for airport operations management exist, and they can provide tremendous help with coordinating the internal and external stakeholders. Very simple solutions can bring collaboration alive. For instance, McGhee Tyson Airport in Knoxville, TN, has a subscription to an instant messaging online application for internal communication. It is possible to create "discussion rooms" with different users, depending on the issues and the stakeholders. External stakeholders can participate as well, and documents can be shared. The same platform—and account—is used by the airport-to-airport mutual aid program, Southeast Airports Disaster Operations Group.

CDM for Smaller Airports

Can smaller airports with limited resources implement ACDM? The answer is yes. It is possible to tailor an ACDM process to fit and specifically address the needs of individual small airports with simple tools and software solutions adapted to their specificities. All aviation facilities can benefit from a more organized collaboration in operations planning and real-time operations, from general aviation to non-hub airports. The case studies of Lake Tahoe Airport, South Lake Tahoe, CA, and North Carolina DOT–Division of Aviation on emergency management detailed in Chapter 2, Operating with ACDM, provide insightful examples of how collaboration at or with smaller airports can benefit these facilities and a broader statewide community of stakeholders.

CDM for Airport Systems

The ACDM approach can apply to multi-airport systems at larger metropolitan areas and state aviation systems. Also, in certain locales, it is going to be needed because different airports of the same region can be interdependent.

For example:

- Dallas/Fort Worth International and the surrounding airports share their available capacities through a digital platform for airport operations in order to smooth massive flight diversions from the larger airport to the smaller facilities. This includes the availability of gates for larger aircraft and Federal Inspection Services for international flights.
- Similarly, in the New York area, all departures from all the airports (Newark Liberty International, LaGuardia, John F. Kennedy International, Teterboro, Westchester County, and others) utilize the same departure fixes. To properly allocate the departure resource, departure demand at these airports is needed by ATC, not just that of the major airline passenger airports. The further in advance this demand is known, the better the allocations can be distributed.
- Southeast Airports Disaster Operations Group, an airport-airport mutual aid program, has ACDM features. Member airports, stakeholders, and professional organizations exchange information on coming adverse events and coordinate their response through an instant messaging platform.

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ACDM and the NAS

The present project focuses on the enhancement of the management of adverse conditions with ACDM. *ACRP Report 137* (Vail et al. 2015) discusses the ATM implication of ACDM and how it relates to the NAS and the current FAA/Industry CDM process. At some point, these different aspects of ACDM and the FAA/Industry CDM meet. The features presented in the toolbox are designed to accommodate advanced flight operations data-sharing initiatives as part of the local ACDM process, as explored in *ACRP Report 137* (Vail et al. 2015).

Emerging ACDM Applications

The impact of other airspace users, such as spacecraft and air mobility network companies, on airport capacity and delay might call in the near future for real-time coordination or at least an exchange of information with airport stakeholders. For instance, in the near future, some ACDM features might be necessary to facilitate the operations of Urban and Rural Air Mobility systems near aviation facilities.

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CHAPTER 2

Operating with ACDM

2.1 ACDM Operations Process

What Is the ACDM Operations Process?

The ACDM operations process is the way the stakeholders of the real-time operations interact to manage adverse conditions. This process is not only for tactical planning ahead of coming adverse situations but also for monitoring the airport status and indicators of foreseeable events during real-time operations.

The ACDM operations process should be as simple and straightforward as possible. It should be described in an ACDM Operations Plan that defines the role of the participants in the process, identifies a leader or coordinator of the overall process in real time, and lists all the applicable plans and procedures.

The local ACDM operations process for managing adverse conditions should be developed by the coalition of the stakeholders of airport operations following the six-step ACDM planning process described in Chapter 3.

Note: The ACDM operations process for enhancing the management of adverse conditions that is proposed in this chapter should not be confused with the ACDM turnaround process for ATM purposes. Nor is it equivalent to the tactical, rolling operations plan often referred to as the Airport Operations Plan (AOP) in the International Civil Aviation Organization (ICAO) literature.

Simplified ACDM Operations Process and the Collaborative Decision Sheet

A seven-step standard process is proposed as the framework of any ACDM operations process (Figure 6). The proposed seven-step process is generic enough to be applied to any airport or group of airports.

The seven steps are as follow:

1. **Gather information and intelligence.** Monitoring airport key performance indicators and reviewing forecasts help with early identification of foreseeable events. A proactive alert of the concerned stakeholders will allow time to prepare, so monitoring and reporting during regular operations are important parts of this process. Means to achieve this monitoring are discussed in Chapter 3.
2. **Consider policies and procedures.** Specific provisions in applicable policies, plans, and procedures should be identified for informing decision-makers. The matrix of available plans and procedures of the ACDM Operations Plan described later in this chapter and the documents featured in the local ACDM toolbox are useful resources for this task.

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22 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

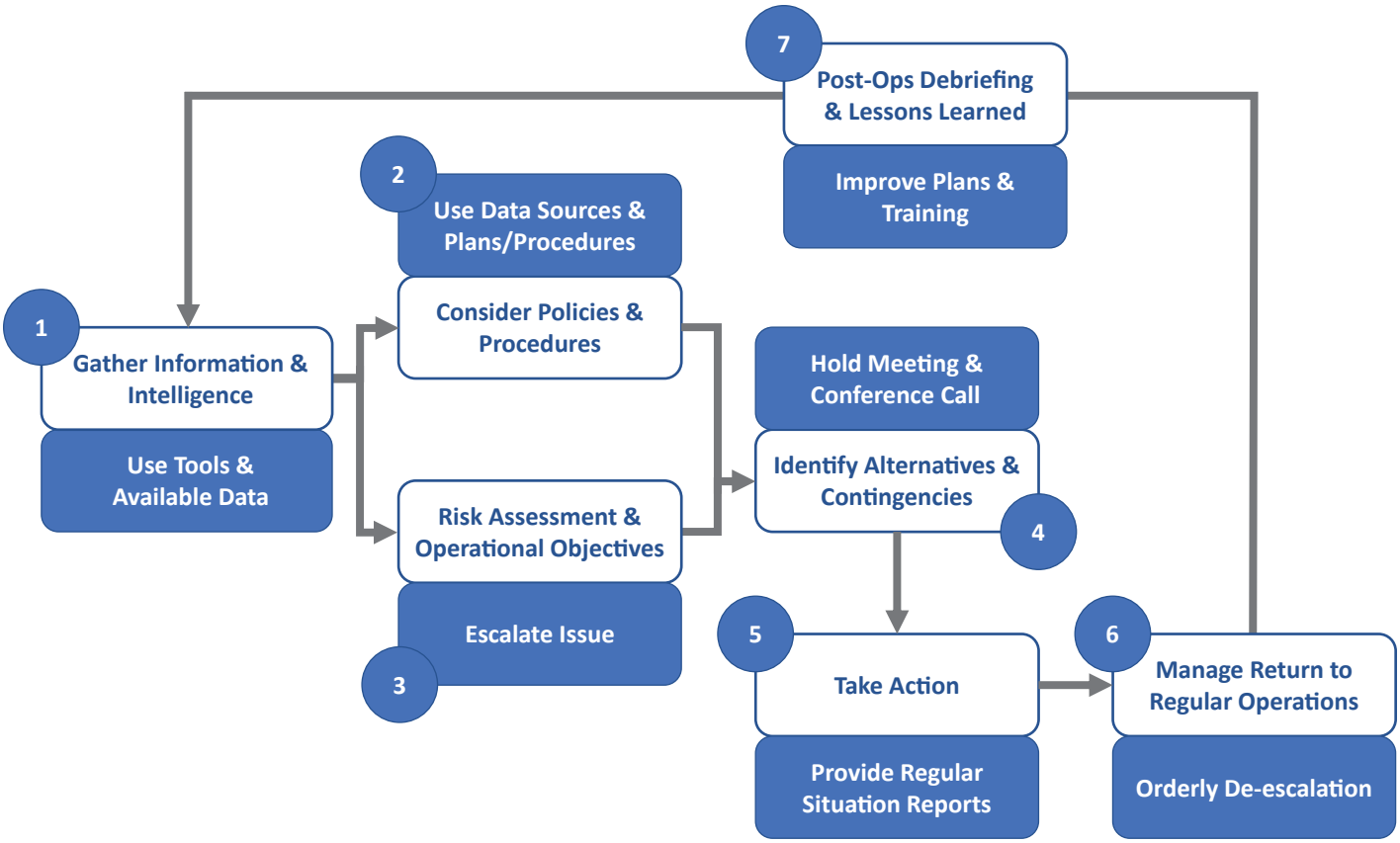


Figure 6. Simplified ACDM operations process.

3. **Assess risk and set operational objectives.** A simplified risk assessment of the coming or ongoing adverse conditions should be made. Risks and opportunities should be discussed and prioritized. Operational objectives should be set: What should be done to contain the effect of these conditions?
4. **Identify alternatives and contingencies.** The coordinator of the ACDM operations process (e.g., the Airport Operations Manager) should convene the stakeholders of real-time operations concerned about the coming or ongoing adverse conditions for a meeting and conference call. During this meeting and conference call, the previous items should be presented, alternatives and contingencies should be discussed, and an event-specific response plan should be prepared and agreed on.
5. **Take action.** Actions to be taken as part of the response should be listed and action leaders should be identified. Regular situation reports should be provided. The frequency and list of recipients of these reports will likely depend on the characteristics of each event. Some events at larger airports might warrant hourly emails to the broader community of stakeholders and the airport upper management. Guidance to the operating staff should be included in the ACDM Operations Plan.
6. **Manage return to regular operations.** The de-escalation should be orderly and the return to regular operations should be managed in order to optimize resources and prevent "side effects" of the response.
7. **Conduct post-operations debriefing.** In the hours or days after the event, a post-operations debriefing should be held to compile lessons learned, improve plans and procedures, and train the teams to enhance the response.

Who should coordinate this process in real time? While the answer to this question might vary at some airports due to local specificities, the airport operator is in the best position for taking this role because of its unique and central role within the operations community. The airport operator

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is at the interface of most, if not all, the stakeholders. More precisely, the Airport Operations Manager or, at smaller airports, the Airport Operations Manager counterpart would be the best choice at most aviation facilities. This provision should be agreed upon among the ACDM stakeholders and mentioned in the ACDM Operations Plan.

A template is provided in the toolbox for supporting decision making based on this seven-step process. The Collaborative Decision Sheet (CDS) is designed to be used as a template and workbook in real-time operations (Figure 7). It should be made available in large format (e.g., reproduced on an 11- × 17-in. printout or engraved on a large whiteboard). The CDS should be used as a common language among the stakeholders and should be archived for lessons learned purposes. The CDS template is available in the "Operate" section of the toolbox, either through the search tool or the easy access menu.

ACDM Operations Plan

The ACDM operations process should be described in an ACDM Operations Plan (ACDM-OP) available to and known by all the stakeholders. The ACDM-OP should address at least the following:

- Purpose of the ACDM-OP,
- Definitions and abbreviations,
- Participants in the ACDM organization (list of stakeholders),
- Description of the ACDM operations process,
- Matrix of responsibility, and
- List of events covered by existing collaborative plans and procedures.

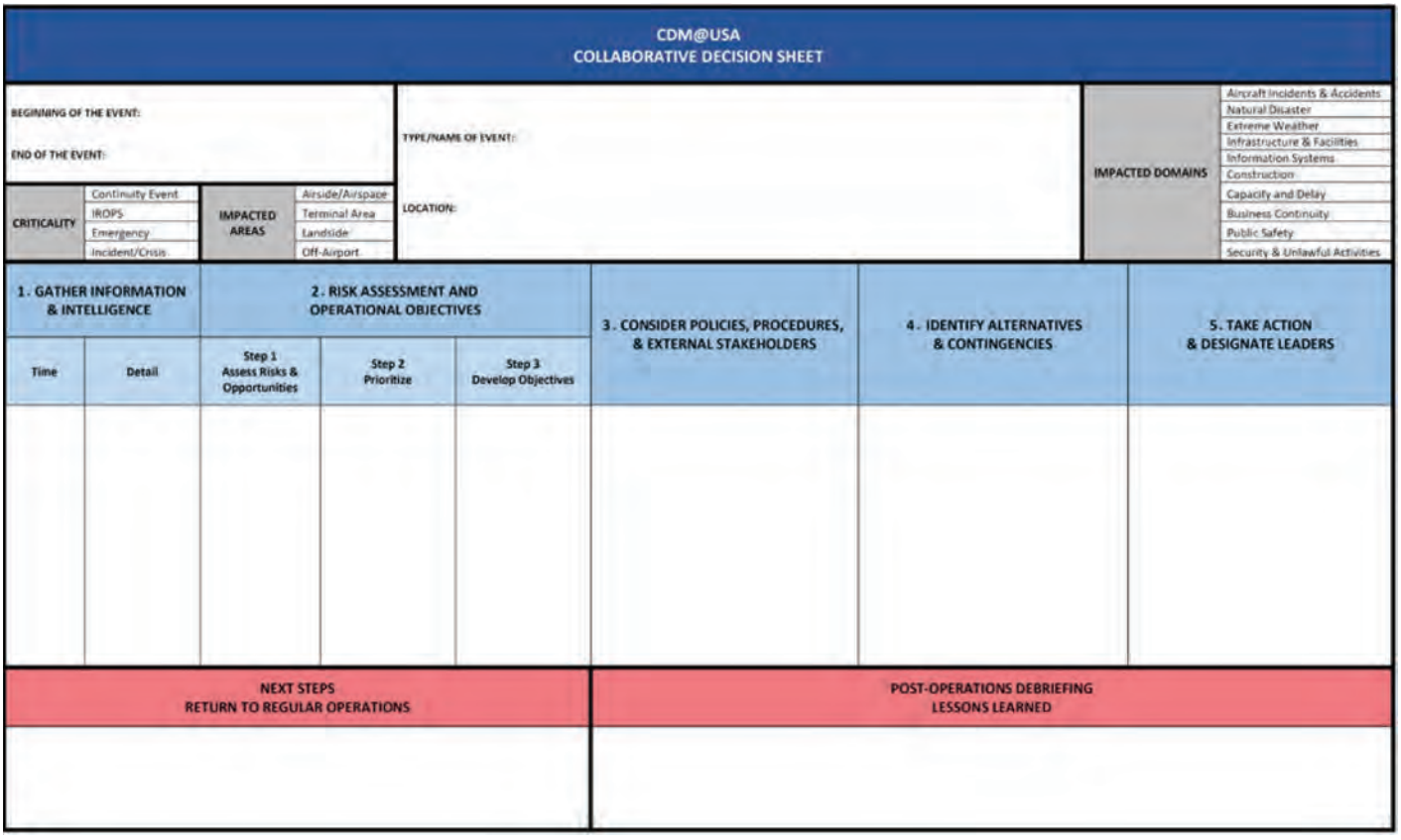


Figure 7. Collaborative Decision Sheet.

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Similar decision-making processes might already exist at the airport for IROPS or emergency operations. In this case, the ACDM operations process would extend the existing IROPS or emergency operations process to events of lower criticality, such as continuity events, and perhaps include domains or areas of the airports that are not yet covered by other plans or processes.

The ACDM operations process should feature different modes depending on the criticality of the adverse conditions. In particular, the ACDM-OP should not conflict with the Airport Emergency Plan (AEP). While the development of the ACDM-OP can lead to revisions of the AEP for efficiency purposes, it is understood that the AEP should take precedence over the ACDM-OP for emergency operations matters.

2.2 ACDM Planning Process

What Is the ACDM Planning Process?

The ACDM planning process is an adverse condition contingency planning process to be used for developing and improving event-specific collaborative plans and procedures. It is built on the six-step process featured in *ACRP Report 65: Guidebook for Airport Irregular Operations (IROPS) Contingency Planning* (Nash et al. 2012) and *ACRP Report 153: Guidebook for IROPS Stakeholder Communication & Coordination* (Cogliandro et al. 2016) and inspired by the Incident Action Planning Process also known as "Planning P." The process presented below originates from the IROPS contingency planning process with adaptations to better serve the ACDM operations process.

ACDM Planning Process

The following six-step standard process is proposed to develop collaborative (ACDM) plans and procedures (Figure 8):

1. **Obtain executive (Steering Group) buy-in.** New plans or procedures are proposed to the Steering Group, which is the executive body of the ACDM implementation project (see Chapter 3). A project group, called the Technical Subgroup, is created for the purpose of preparing

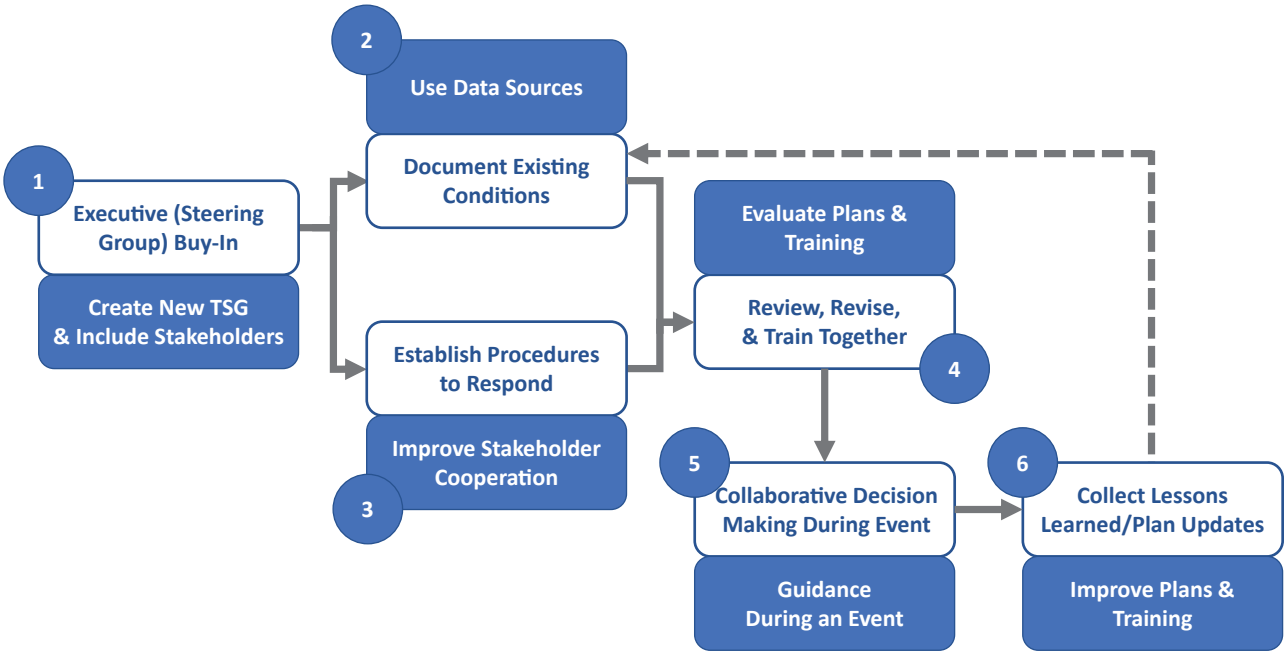


Figure 8. Six-Step ACDM planning process.

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the document, and it should involve all the stakeholders concerned about or impacted by the adverse event or conditions that trigger the initiative.

2. **Document existing conditions.** Stakeholders identify and gather relevant response plan information to ensure that proper communication, cooperation, and coordination occur. The results of this review should include both formal and informal coordination among these organizations. Response plans from individual organizations should be evaluated for adequacy.
3. **Establish response procedures.** A procedure is developed for responding to the specific adverse event/conditions. The procedure should be simple and straightforward but at the same time provide enough guidance for the stakeholders to organize, communicate, and address the issue. The procedure should describe the type of event covered; describe the roles of each stakeholder and the resources involved; provide, as far as practicable, a step-by-step response process; include a communication plan; address known or potential operational issues; and recommend mitigation steps.
4. **Review, revise, and train together.** The procedure is reviewed and revised as needed. Training should be considered for operational readiness purposes.
5. **Apply CDM during an event.** The procedure is considered part of the seven-step ACDM operations process (see Chapter 1) and is applied collaboratively.
6. **Collect lessons learned and plan updates.** A formal lessons-learned meeting is held. The last step of the ACDM operations process can serve this purpose. The procedure should be improved based on these lessons learned.

2.3 Impact on Current Practices

Purpose

This section explores the impact of ACDM on current practices in airport operations. This analysis includes real-time coordination among the stakeholders, IROPS, emergency management, and Safety Management Systems for airports.

Real-Time Coordination Among Stakeholders

ACDM aims to enhance the management of airport operations among the stakeholders through a collaborative framework. It is expected that the ACDM operations process will dramatically strengthen the relationship and the coordination among the stakeholders of real-time airport operations. This process should remain the same from one airport to another as it is a broad, generic process applicable to any aviation facility or system. Airports might implement daily calls and reports with the operations community.

The ACDM operations process, the systematic use of the CDS, and the regular coordination among the stakeholders promoted in the toolbox may add a limited workload to the airport operations staff. However, the benefits of this process will greatly exceed the costs. Costs include the time spent in training, holding one or more daily phone briefings, populating the CDS for decision-making purposes, and performing post-operation debriefings for lessons learned purposes.

Irregular Operations Contingency Planning

While preparing plans and procedures requires time, the ACDM planning process itself is not expecting to adversely affect the current organizations and practices. This process should save time and enhance the efficiency of the contingency planning effort. The proposed ACDM planning process is derived from the IROPS contingency planning process addressed in *ACRP Reports 65* (Nash et al. 2012) and *153* (Cogliandro et al. 2016).

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Airports already utilizing the IROPS planning process can easily integrate it into their ACDM initiative. The ACDM organization and process should encompass any IROPS initiative already in place.

The IROPS Contingency Response Committee could become a Technical Subgroup of the ACDM organization. Its mandate and that of the IROPS Champion within the local ACDM could be broader and include collaborative contingency procedures. Generally speaking, the ACDM implementation should create opportunities and a framework for expanding the current IROPS contingency planning practices to events of lower criticality (e.g., continuity events) that are not necessarily covered by plans and procedures. Existing IROPS contingency planning practices will stay mostly untouched by the ACDM initiatives.

Emergency Planning and Management

The implementation of ACDM should not adversely affect the emergency management structure already in place at airports. However, ACDM can contribute to the enhancement of emergency planning and facilitate the response in different ways. First, emergency planning can benefit from the proposed process for ACDM planning. Then, the collaborative environment created by the ACDM initiative can facilitate the discussions among the stakeholders of emergency management—when planning for events and when responding to them. Finally, airports might elect the enhancement of emergency management as one of their priorities for the ACDM community as a whole and make it a strategic goal of the ACDM initiative (see Chapter 3, Implementing ACDM). They can choose to feature specific emergency planning and management in their implementation roadmap. ACDM for emergency planning and management provides leverage that can help give visibility to this domain and connect it to non-traditional stakeholders, direct resources to it, and create a favorable collaborative environment that can be an effective multiplier when providing a response to an emergency situation.

The ACDM initiative can also be used for facilitating the collaboration between airports and the local emergency management agencies for planning and readiness purposes. Existing best practices in this domain are available in *ACRP Synthesis 50: Effective Cooperation Among Airports and Local and Regional Emergency Management Agencies for Disaster Preparedness and Response* (Smith 2014) and *ACRP Synthesis 73: Emergency Communications Planning for Airports* (Smith et al. 2016).

Safety Management Systems for Airports

While they both rely on collaboration and transparency and are systematic and systemic approaches to their domain of airport operations, Safety Management Systems (SMS) and ACDM have different focuses. SMS are systems and processes focusing on aviation safety. ACDM focuses on throughput. They do not supersede each other or interact directly.

However, ACDM and SMS are mutually beneficial in the way that they both contribute to a work environment, favoring collaboration and transparency across the stakeholders. Also, the data gathered as part of the ACDM process may be occasionally beneficial to safety studies and incident analysis performed under the airport SMS.

Finally, several plans, procedures, systems, and other projects developed as part of the ACDM implementation roadmap might affect airport operations and the infrastructure. Some of the changes created by these might warrant Safety Risk Assessment and a Safety Risk Management Plans as part of the local airport SMS.

More information on SMS for airports is available in the FAA Draft Advisory Circular 150/5200-37A (FAA 2016), FAA Order 5200.11: *FAA Airports (ARP) Safety Management System* (FAA 2010), as well as *ACRP Report 1: Safety Management Systems for Airports* (Ludwig et al. 2007).

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Airport-Specific CDM Projects

Each airport will decide to bring under the local ACDM initiative different projects interfacing with different stakeholders, enhancing the management of adverse conditions or improving the throughput. While the ACDM process should be the same across the NAS, the ACDM implementation described in the "Implement" section of the toolbox might be accompanied by physical projects in support facilities (e.g., Airport and Emergency Operations Centers), information systems (e.g., airport operations solutions), and infrastructure (e.g., airfield enhancements and navigational aids) that will be mostly airport specific.

The impact of these projects on the workload of the operating staff and the potential need for additional resources should be considered when evaluating their feasibility and comparing their costs and benefits. Guidance on these matters is provided in Chapter 3. However, these impacts are not ACDM specific. While the ACDM initiative might help in coordinating these projects and, at some point, facilitate their emergence, they are not necessarily directly related to the ACDM operations process.

Summary of Impacts

Table 3 summarizes the expected impact of ACDM implementation on existing airport practices.

Note on ACDM Implementation Resources

While it is not expected that the ACDM operations and planning processes alone will have an impact on organizations, ACDM implementation might monopolize a significant part of the time of some of the project managers at the ACDM project management office. At larger airports, the ACDM initiative typically requires a full-time CDM Manager. Its deputies—the CDM Champions—are usually not full-time CDM contributors. The ACDM implementation, like that of any project/program, might require additional resources, depending on the ambitions of the local ACDM community and the individual projects that may be brought under the umbrella of the ACDM implementation roadmap. These resources should be considered multipliers that will increase the benefits of these individual projects and ensure they are implemented in a smooth and timely manner.

Table 3. Summary of the impact of ACDM implementation.

Domain	Overall Impact of ACDM	Opportunities
Real-Time Coordination	Medium	+++
IROPS	Medium	+++
Emergency Management	Low	+++
SMS for Airports	Low	+
Airport-Specific Projects	Variable	Variable

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CHAPTER 3

Implementing ACDM

3.1 Fundamentals of ACDM Implementation

What Is ACDM Implementation?

ACDM is a process and also a state of mind of stakeholders agreeing to work collaboratively—not side by side—in operations planning and real-time operations. Switching from a pre-ACDM organization to an ACDM one requires time to convene the stakeholders, design the local ACDM process, decide on the resources to dedicate, and educate the operations community. This is why implementation is such an important aspect of a successful ACDM process. It typically requires a specific project management structure, which will ultimately manage the local CDM, and a realistic "implementation roadmap" to prioritize the different projects brought under the ACDM initiative.

ACDM Starts with a Team

ACDM requires strong executive buy-in and support from the different stakeholders. This support is of strategic importance because it sets up strengthened operations and overall airport resilience. Even though one of the stakeholders should lead and manage the ACDM initiative, all internal and external stakeholders should embrace it and contribute to its success. ACDM cannot be a one-party journey.

It is crucial that operating staffs understand the benefits of ACDM and adopt it, because they are the ones who make ACDM work and keep it alive. More than a process, ACDM is a philosophy and is about agreeing to work together transparently. To achieve this, a strong bottom-up component should be established through workgroups and conferences during which the operations community can contribute to shaping the local ACDM, provide feedback, and share lessons learned.

A Joint Vision with Shared Ambitions

After agreeing on general terms for collaboration, the stakeholders should gather to define the following:

- **Strategic goals**. These are high-level directions for enhancing local operations. They address the following questions: Why do we want to implement ACDM locally? How can we work together to improve our operations and make them more resilient?
- **Operational objectives**. These are short-term objectives—to be met within a 5-year horizon—that translate the strategic objectives into practical and implementable targets that are going to structure the ACDM initiative. They address the following question: What should we do first to achieve the strategic objectives?

The local ACDM should reflect the joint vision of the stakeholders and address shared ambitions. The strategic goals should be defined by the local (airport) decision-makers representing

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the stakeholders. The operational objectives should be proposed by the stakeholders' operations leaders. The goals and objectives should be unanimously adopted and recognized by the airport operations community as a way to enhance operations and make them more resilient.

Roadmap CDM@USA: The ACDM Program

Achieving the operational objectives and, more generally, implementing ACDM requires setting new plans and procedures, making some organizational changes, and perhaps developing or procuring new systems and equipment. All these improvements are projects for the wider ACDM implementation program or "roadmap" to be completed at the previously defined horizon. This local ACDM implementation roadmap should, as far as practicable, identify projects specific to individual stakeholders that could positively or negatively affect the other stakeholders' projects. The size, extent, and complexity of the ACDM implementation roadmap will depend on the airport specificities (e.g., size, traffic, stakeholders, and issues to be addressed).

Working Together for the Success of ACDM

Implementing ACDM and completing all the projects of the ACDM implementation roadmap calls for a project/program management organization. The members of this project management group or "office" are not necessarily utilizing 100% of their time working on the CDM implementation, except perhaps at some large-hub airports with comprehensive and complex roadmaps. The project management group/office should represent the diversity of the stakeholders in order to ensure the participation of their respective entities in the ACDM initiative and promote ACDM internally. These project team members are called CDM Champions, with one leading the group as CDM Manager.

Change Management Aspects

ACDM has change management aspects that should not be neglected. Creating new missions for CDM Champions, promoting deep collaboration with external stakeholders, and setting up new procedures and perhaps new systems and equipment are the many modifications of the operating conditions and working environment that can adversely affect individual and collective performance if they are not carefully planned. Impacts on the organization and the workforce should be evaluated, and the human factor should be considered. The operations team members should be trained and prepared for these changes through an ACDM curriculum adapted to their role within the ACDM process and organization.

Standardization and Certification

Again, a local ACDM process should serve the needs and the vision of the airport where it is being implemented. However, although each airport is different, no airport is completely independent and isolated from the others, and they are ultimately all part of the U.S. NAS. It is important that we speak the same language to ensure interoperability and facilitate information and experience sharing. ACDM should not create walls inside the NAS; it is intended to create bridges in and between aviation facilities and stakeholders.

The guidance proposed in the toolbox provides flexibility to adapt ACDM to all airports and airport systems regardless of their level of traffic, location, and other specificities. However, it promotes standardization through a common language, unique step-by-step processes, and a self-certification sheet. Airports pursuing ACDM implementation should follow these directions as far as practicable to enable interoperability and mutual understanding from airport to airport.

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3.2 Stakeholder Identification and Executive Buy-In

Who Should Initiate the Local ACDM Initiative?

ACDM is typically initiated by a small group of stakeholders involved in operations efficiency and resilience. These stakeholders typically include the airport operator, the ATCT, and the main air carriers, because ACDM usually starts from the flight operations and airside/airspace.

The airport operator is generally in the best position to coordinate the impetus for all discussions on the ACDM initiative. This is because the role and mission of the airport operator encompass or interface with all the different domains and functions of an airport from the curbside to the airspace, and that of the large variety of stakeholders from transit agencies and city/county departments to the different branches of FAA.

The notion of the airport operator is evident most of the time because most airports are controlled and operated by local governments (i.e., cities and counties) and sometimes even states. However, a growing number of airports (e.g., Chicago Midway International Airport, Ontario International Airport, Orlando Sanford International Airport, and New York Stewart International Airport) have entered into agreements (e.g., partnerships, management-for-fee, and outsourcing) with private firms for part or all of the management services. In those cases, the leading entity should carefully choose among the main stakeholders.

Who Should Be Part of the Local ACDM Initiative?

The short answer to this question is, "Everyone who has an impact on airport capacity and resilience," because ACDM has the ultimate ambition to federate all the stakeholders of airport operations on these two matters. However, stakeholder involvement can vary. A successful ACDM project needs to acknowledge that the participation level in the implementation project and the ACDM process may differ among the stakeholders.

Stakeholder Identification

Stakeholders potentially involved with ACDM can significantly vary from one airport to another. The following inventory proposes an overview of the different stakeholder groups and their members who are critical to the ACDM process or bring a specific value-added effect (Table 4).

Executive Buy-In: The Top-Down Component of ACDM

Executive buy-in is essential to the success of the ACDM initiative, both for its implementation and its continuation once in place. The executive management of the ACDM signatories must have a firm commitment to develop, implement, and continually improve the ACDM process. They must also promote ACDM internally as a priority for improving operations efficiency and resilience in order to illustrate a full top-down buy-in within the airport staff.

Memorandum of Understanding

The Memorandum of Understanding (MoU) is the birth certificate of the local ACDM initiative. It creates a general, high-level framework that will be the vehicle of the ACDM implementation project. An MoU is a high-level agreement that formalizes a convergence of will among parties, indicating an intended common line of action. The MoU should include the following:

- **Description of the ACDM.** This is a general description of what ACDM is and its expected benefits to each stakeholder or category of stakeholders.

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Table 4. Inventory of potential ACDM stakeholders.

Stakeholder Group	Definition and Roles	Example(s)
Airport Operator	This category includes the internal stakeholders of the airport operator concerned with airport operations.	▪ Aviation services ▪ Airport operations ▪ Communications ▪ Emergency management ▪ Engineering and maintenance ▪ Public safety ▪ Terminal operations
Airport Facility Operators	Many airports have terminals and other facilities operated by third parties. This category includes fixed-base operators, passenger terminal facilities operated by third parties (e.g., Propeller Airports at PAE, JFKIAT at JFK Airport's Terminal 4, Lonestar Airport Holdings at AUS, and Private Suite at LAX), and contractors providing critical management services.	▪ Terminal operators ▪ Fixed-base operators ▪ Parking garage operators
Air Navigation Service Provider	The FAA is the air navigation service provider within the U.S. National Airspace System. However, some air traffic control facilities are staffed by private companies under contract with FAA.	▪ FAA ATCT ▪ FAA Terminal Radar Control ▪ FAA TechOps ▪ Federal control tower operator
Apron Management or Ramp Control Service Providers	Apron management service providers include airline hub control centers; all entities providing ramp control services, gate planning, and resource allocation; and other apron management- or ramp control-related services.	▪ Hub control center of the main air carrier ▪ Ramp towers
Flight Operators	Flight operators are a broad category that includes but is not limited to air carriers. Urban/Rural Air Mobility operators fall under this category.	▪ Air carriers ▪ Air taxis/urban air mobility ▪ Air ambulances and aerial work services ▪ Commuters
Aircraft Ground Support Providers	Aircraft ground support includes a wide variety of servicing activities directed toward the aircraft on the ground. At airports regularly facing adverse winter conditions, the de-icing service providers should be ACDM partners.	▪ Aircraft de-icing service providers ▪ Aircraft fueling ▪ Catering ▪ Multi-service ground handlers
Other Aviation Tenants	This category includes any other aviation tenants that do not fall under one of the previous categories and that could have a role in the ACDM process.	▪ Aircraft recycling center ▪ Maintenance repair and overhaul centers
Ground Transportation (Landside)/Ground Mobility Service Providers	This category includes all entities providing (ground) mobility services from and to the airport.	▪ Public transit agencies ▪ Taxi companies ▪ Transportation network companies ▪ Water taxis or ferries
Hotels	Hotels are important stakeholders involved in accommodating passengers on flights being diverted or canceled, as well as providing lodging to airport workers mobilized for providing a response to prolonged adverse conditions (e.g., snow removal).	▪ On-airport and selected off-airport hotel facilities
Emergency Management, Public Safety, and Law Enforcement	The airport police, paramedics, aircraft rescue and firefighting (ARFF), and state and local public safety, emergency management, and law enforcement bodies should be part of the ACDM initiative.	▪ Airport police ▪ Airport emergency medical care ▪ ARFF ▪ DHS (CBP and TSA) ▪ Search and rescue services ▪ State/local emergency management agencies ▪ Other jurisdictions

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32 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

- **Objectives of the MoU.** This section describes the intentions of the memorandum. MoU objectives typically include:
  - Creating the cooperative framework to implement ACDM.
  - Ensuring technical mechanisms allowing common information sharing.
  - Implementing procedures to increase traffic predictability and operations resilience.
  - Setting up monitoring mechanisms in order to evaluate efficiency and resilience and propose further optimization.
- **Commitment of the signatories.** This is a "moral engagement" of the signatories to support the local ACDM initiative, dedicate resources to this effort, and follow the collaborative procedures. Typical MoU commitments are the following:
  - Ensure active participation in all levels and phases of the implementation project as required.
  - Support the development and validation of procedures or systems part of the ACDM implementation roadmap.
  - Follow the collaborative ACDM plans, operational procedures, and rules.
  - Share information under the agreed conditions and act on the shared information.
- **ACDM organization.** This section briefly describes the main elements of the ACDM organization. These elements are further described in section 3.3, ACDM Organization. The main bodies are:
  - Steering Group (SG) that provides executive leadership to the initiative,
  - Working Group (WG) led by the CDM Manager and attended by the CDM Champions, and
  - Technical Subgroups (TSGs) that conduct specific discussions or projects.
- **General provisions on costs.** The principle is that costs associated with equipment or resources will be covered by each concerned party. This should also apply to any system adaptation or integration unless otherwise approved. Where an interface is required between parties, each one will try to minimize the cost impact on the other.
- **Confidentiality.** This section features general provisions on confidentiality of information exchanged as part of the ACDM initiative.
- **Dispute resolution.** This section explains how disputes should be handled within the ACDM organization on the conduct of the initiative or between members of the ACDM initiative.
- **Amendments.** This section features general provisions of the mechanism for submitting amendment proposals to the SG. The proposed template (see section 3.3, below), suggests referring to the Terms of Reference of the SG, which should be a separate document.
- **Signatures of the ACDM members.** The MoU should be signed by members of the upper management of each signatory who has decision authority (e.g., the airport Chief Executive Officer, the Director of Operations of the air carriers, the Chief Executive Officer or Managing Director of the ground handling service providers, and others). This level of signature will also confirm the strong executive buy-in that is needed to support the adoption of ACDM by the whole community of operations.

The ACDM MoU should be a high-level agreement among the participants in the ACDM initiative. It should not be confused with the more detailed terms of reference setting up the rules of the ACDM workgroups or with any service-level agreement on data sharing. An MoU template is available in the toolbox.

3.3 ACDM Organization

Structure of the ACDM Organization

The ACDM initiative needs a team, and this team should include at least (1) a decision level composed of executive managers, (2) a project/program management level that is the backbone of the ACDM initiative, and (3) project groups combining technical experts and operations leaders who run the different projects that broadly implement ACDM (Figure 9).

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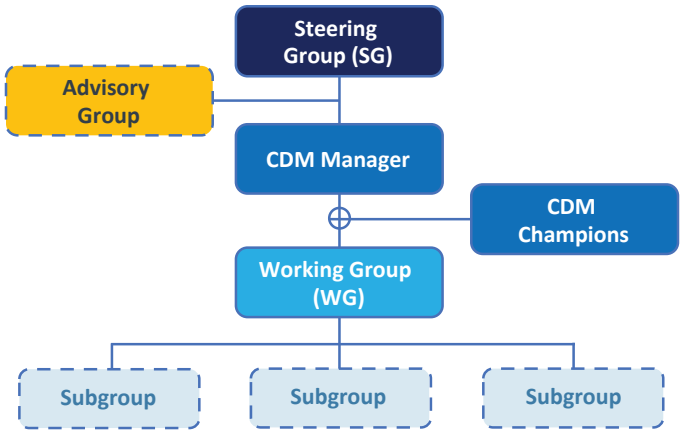


Figure 9. ACDM organization structure.

These three stages correspond to the following bodies:

1. The SG in charge of overseeing the ACDM initiative,
2. The WG providing project/program leadership to the ACDM initiative, and
3. The TSGs created by the WG for performing individual projects.

Steering Group

The SG is the oversight and decisional body of the ACDM organization. The SG should have one representative for each signatory. These representatives should be from high-level management, preferably the chief operations officers of the individual partners or Directors of Surface Operations for airlines. The SG should have access to airport performance reviews. It can decide to create an Advisory Group composed of airport operations experts to provide independent expertise and recommendations through the CDM Manager.

Terms of Reference should be set for the SG. This document should detail the rules of the SG, including membership, mission statement, frequency of regular meetings, and mechanisms to convene extraordinary sessions.

Note: At larger airports, signatories who cannot attend the SG meeting regularly could decide on a group representative (e.g., through the local Airline Operations Committee).

CDM Manager

The CDM Manager is the project/program manager of the ACDM initiative appointed by the SG, responsible for implementing ACDM at the airport. While the CDM Manager might be a full-time employee of one of the ACDM patrons, it is important that the CDM Manager acts for the sake of the ACDM initiative and in the interest of all the stakeholders. The CDM Manager reports to the SG. At larger airports, making the CDM Manager position a full-time job might be required in order to achieve the set objectives and complete on time all the critical projects needed to set up the local ACDM.

Working Group

The WG typically comprises the CDM Manager and the CDM Champions. TSG leaders and project managers may participate by providing the WG with progress reports on their projects.

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The CDM Champion positions are typically not full time. A CDM Champion should be designated within the organization of each stakeholder. CDM Champions act like assistant project managers to the CDM Manager. They supervise the participation of their entity in the ACDM initiative, promote ACDM internally, and regularly inform their management of the progress of the ACDM initiative. CDM Champions perform key roles because they, along with the CDM Manager, keep the ACDM initiative alive.

Technical Subgroups

As explained later in this chapter, ACDM implementation is a program made of many projects—mainly in operations and information systems. All these projects cannot necessarily be managed by the CDM Manager and the CDM Champions. Additional resources might be required. Consequently, TSGs should be created for driving individual or group projects. A leader should be chosen for each TSG who should send regular project progress reports to the CDM Manager.

TSGs can be created for incorporating existing collaborative initiatives within the ACDM organization. For instance, the airport IROPS Contingency Response Committee and the "Snow Committees" could become TSGs of the ACDM organization.

3.4 Defining the ACDM Project

Defining Common Goals and Objectives

After agreeing on general terms of collaboration, the stakeholders should gather to define the following:

- **Strategic goals.** High-level goals for the desired end-state of airport operations. They address the questions: Why do we want to implement ACDM locally? How can we improve operations efficiency and resilience together?
- **Operational objectives.** Short-term (up to 5-year) objectives translating the strategic objectives into practical, implementable targets that are going to structure the ACDM initiative. They address the question: What should we do first to achieve the strategic objectives?

Strategic Goals

The local ACDM project should reflect the joint vision of the stakeholders and address shared ambitions. The strategic goals should be defined by the SG, while the operational objectives should be proposed by the WG. The strategic goals should be unanimously adopted and recognized by the airport operations community as a way to enhance operations and make them more resilient.

The strategic goals should be defined during an SG workshop that is moderated by the newly appointed CDM Manager. The members should ask themselves the following questions:

- Why do we want to implement ACDM at our airport?
- How can we work together to improve operations efficiency and resilience?

Leaders should be invited to make high-level observations on the existing hindrances to operations efficiency and resilience. The following observations—inspired by issues faced by real airports—illustrate the expected final statements from the SG members:

- **Observation 1.** Internal and external stakeholders should collaborate better in real time. We talk to each other only when there is a significant issue. There is no broad vision of the airport resource and capacity shared in real time throughout the airport operations community.

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- **Observation 2.** Several hindrances can be easily removed to significantly improve efficiency overall. The SG has identified specific issues to be addressed collaboratively [attach list] as part of the ACDM implementation.
- **Observation 3.** Gate resource is a significant bottleneck that should be improved. Besides the addition of new gates, progress can be made on the tactical and real-time management of the gate resource. Aircraft wait too long on the taxi lanes. Marshallers and wing walkers should be ready when the aircraft arrives, even when the flight is early/late. Pushback should be on time as well.
- **Observation 4.** Recent adverse events could have been better managed with joint procedures. Stakeholders do not necessarily know what others know and what others can or cannot do for them. There is a lack of mutual understanding on many issues.
- **Observation 5.** The winter operations staff recommends more training before each season and suggests inviting airline representatives and the ATC staff. Formal debriefing sessions with lessons learned should be held after each winter season in order to improve our performance over time.
- **Observation 6.** Each stakeholder has a piece of critical information that is not necessarily available to the others. The airport operator usually learns about diverted flights when they are about to land. The ATC staff have little information about the aircraft de-icing capacity. Airlines would like to get more information on airport resources during construction.

The strategic goals should derive from these observations. It is not required that each observation leads to a strategic objective. Instead, observations should be grouped together as far as practicable to limit the number of strategic objectives to around 5.

- **Observation 1** → *Strategic Goal 1:* Integrate real-time operations management.
- **Observation 2** → *Strategic Goal 2:* Enhance operations efficiency overall.
- **Observation 3** → *Strategic Goal 3:* Enhance gate resource management.
- **Observations 4 and 5** → *Strategic Goal 4:* Enhance adverse conditions readiness.
- **Observation 6** → *Strategic Goal 5:* Share real-time operations data.

A date (year) should be agreed on to achieve these strategic goals. An ACDM implementation roadmap typically covers a 5-year timeframe. Five years is also a reasonably foreseeable horizon for improving operations, and it can be aligned with the investment plans of the stakeholders as needed.

Each airport is different, and so different operational deficiencies or difficulties can be identified as priorities to be addressed through enhanced collaboration. For instance, "expediting snow removal on the non-movement area" could be among the operational objectives of an ACDM initiative at an individual airport in Washington State, but will most likely not be a priority in Florida. In two comparable airports of the same region, one can consider gate management in general as a weakness when another will focus on adverse weather conditions. Consequently, it is not expected that all airports will pursue the same strategic goals and operational objectives.

Operational Objectives

The WG should work on operational objectives that will translate the strategic goals into short-term, practical targets. Each strategic goal should have at least one operational objective to be completed by the end of the ACDM implementation roadmap horizon. However, it is possible for a strategic goal to lead to several operational objectives with different deadlines.

For instance, the fictional airport used in the example could decide to create three different operational objectives (OOs) for achieving the example Strategic Goal 1:

- **OO1.** Provide collaborative tools to airport and stakeholders by 2022.
- **OO2.** Create a new Integrated Airport Operations Center by 2023.
- **OO3.** Integrate ACDM patrons to the Airport Operations Center by 2025.

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The operational objectives proposed by the WG should be reviewed and approved by the SG. A first rough estimate of the resources needed to achieve the local ACDM objectives can typically be prepared at this point.

Roadmap CDM@USA: The ACDM Program

Achieving the operational objectives, and more generally implementing ACDM, requires setting up new plans and procedures, making some organizational changes, and perhaps developing or procuring new systems and equipment. All of these improvements are the many projects of the wider ACDM implementation program or "roadmap" to be completed at the previously defined horizon.

Based on the set operational objectives, the stakeholders should convene during a special ACDM workshop for drafting an inventory of all the enhancements needed. All together, they make the ACDM implementation roadmap. For instance, building on the three operational objectives (OO) of the previous examples, individual projects (P) at this fictional airport could be:

- **OO1.** Provide collaborative tools to airport and stakeholders by 2022
  - **P1.** ACDM portal sharing flight operations data and real-time key performance indicators
  - **P2.** Real-time chat rooms for the stakeholders of airport operations
  - **P3.** "Hotline" between the ATCT and the Airport Operations Center
- **OO2.** Create a new Integrated Airport Operations Center by 2023
  - **P4.** Integrated Airport Operations Center (terminal and airside)
  - **P5.** Migration of Airport Security to Integrated Airport Operations Center
- **OO3.** Integrate ACDM patrons to the Integrated Airport Operations Center by 2025
  - **P6.** Wanna Fly Airlines Hub Control Center liaison at the Integrated Airport Operations Center
  - **P7.** ATCT liaison at the Integrated Airport Operations Center

It is understood that a specific project might address or contribute to addressing more than one operational objective. Projects should be located on the ACDM implementation timeline. A precise schedule is not needed at the end of the first ACDM workshop. The main goal of the workshop is to identify the needs and to place them on the timeline for further planning effort. Some projects might require the approval of the SG and individual stakeholders because of the resources needed.

The ACDM implementation roadmap should integrate or mention projects that, though not necessarily part of the ACDM project or common to different stakeholders, could affect the community of airport operations' efficiency and resilience. Major capital improvement projects, such as the construction of a new terminal or end-around taxiways, should be mentioned on the ACDM implementation roadmap even if they are handled by a separate and specific project management structure. This structure should coordinate and inform the WG, though massive capital improvement projects can warrant modifications to the ACDM implementation roadmap. The ACDM organization can address or facilitate certain issues faced by the capital improvement projects.

ACDM Implementation Workbook

An ACDM Implementation Workbook is proposed to support the initial steps described in this chapter. The workbook should be shared within the ACDM organization and beyond as necessary to inform the stakeholders, to enable executive buy-in, and to inform all parties about the nascent ACDM initiative.

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3.5 ACDM Implementation in Motion

After Drafting the ACDM Implementation Roadmap

After drafting the local ACDM implementation roadmap, an estimate of the budget, an identification of project funding, and an inventory of the resources needed should be prepared and discussed by the WG and then the SG. The ACDM implementation roadmap should be an achievable program that serves the strategic objectives and achieves the operational objectives defined for the ACDM initiative. Improving operations and teamwork takes time, and realistic due dates should be set. The completion of major projects and intermediate operational objectives are the many implementation milestones. Multiple projects might be linked together to form a major sequence of milestones that can be prioritized as "fast tracks" helping to achieve the goals and objectives. The ACDM implementation roadmap should be managed as a program.

Monitoring Each Project

Each project should be monitored by the WG. TSGs and project managers should send regular progress reports to the CDM Manager. Progress reports can and should be as simple and straightforward as possible and should include the following:

- A summary of the work accomplished since the last progress report,
- A list of existing and potential hindrances identified,
- The overall percentage of physical completion of each task, and
- Status update on the funds expended to date.

Other key performance indicators can be added to align with the project management practices of the airport and its stakeholders. However, the bureaucracy should be minimized to focus the effort of the participants on the implementation itself and the delivery of the projects.

The WG should have regular meetings to review these progress reports, to adjust the schedule of the ACDM implementation roadmap and request additional resources to the SG if needed.

Progress Reports on ACDM Implementation

The SG should convene regularly, and the CDM Manager should provide the SG with a progress report on the implementation of the ACDM implementation roadmap. While it might not be relevant to review each individual project with the SG, the major pieces of the roadmap (e.g., fast tracks) might deserve the attention of the members.

The SG should be made aware of the main issues faced by the implementation effort, especially when these obstacles are procedural or cultural and stakeholder specific. The members of the SG should do their best to support and help the CDM Manager succeed and achieve the goals and objectives that they set for the ACDM initiative.

ACDM Newsletter

It is a good practice to have a newsletter to communicate to the community of airport operators about the progress made on ACDM implementation and the success of the different projects. Milestone achievements should be celebrated. Collective successes involving multiple stakeholders should be mentioned as well. Contributors to the ACDM initiative should be put in the spotlight. Email-based newsletters are a simple and efficient way to provide updates. More elaborate newsletters might be prepared at larger airports.

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CHAPTER 4

Featured ACDM Projects

The purpose of this chapter is not to promote specific airport operations enhancement projects over others but to provide typical examples and case studies of successful ACDM improvements. Several of these improvements are discussed in detail in the "Operate" section of the toolbox. Additional resources are proposed in the eLibrary of the toolbox. For some—but not all—of the issues mentioned below, collaborative tools and templates have been developed. When mentioned below, the tool is available in the eLibrary and the "Operate" section of the toolbox, and training materials can be downloaded from the "Train" section of the toolbox.

4.1 Defining Key Performance Indicators

Defining key performance indicators (KPIs) agreed upon among the stakeholders of airport operations for monitoring and decision-making purposes is typically a fundamental element of an ACDM implementation roadmap. This will facilitate discussions on other ACDM projects and real-time operations as well, by establishing a common language throughout the operations community.

These KPIs should all describe the performance of the airport. Below are examples of a simplified set of KPIs for a commercial service airport:

- Airport capacity (including ATC rates),
- Throughput (departures and arrivals per hour),
- Taxi-in and taxi-out times (min/flight),
- Departure queue hold time (min/flight),
- Adherence to earliest off-block time (EOBT) (min/flight),
- Gate occupancy (percentage of contact stands),
- Remote parking occupancy (percentage of total remote overnight parking capacity),
- TSA and Customs and Border Protection checkpoint waiting times (min/passenger),
- Bag claim waiting time (min/bag), and
- Congestion level of ground access and curbside(s) (level of service).

Other KPIs such as the average aircraft de-icing time or the adherence of the actual off-block time (AOBT) to the EOBT can be added, as long as these KPIs are accessible. The approved KPIs should be monitored, as far as practicable in real time, and communicated to the community of airport operations—typically through an application or a website.

The definition of KPIs can be an opportunity to redefine the local notions of capacity and throughput and discuss the means to monitor flows (e.g., boarding pass scan and collection of these times along the passenger journey).

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Such an approach applies to smaller airports (e.g., gathering an exact count of aircraft operations at non-towered airports) and statewide airport systems (e.g., applying the dashboard on the availability of aviation facilities for emergency management purposes, available in the toolbox).

4.2 Daily Coordination Calls and Airport Status Reports

The survey conducted as part of this ACRP project reveals that most airports—including commercial service facilities—do not hold regular calls with the stakeholders of airport operations. Such daily coordination can significantly improve collaboration and readiness by raising awareness on ongoing issues and sharing information on the coming operations. It also contributes to team-building across the operations community by increasing the frequency of direct contacts among its individual members.

A daily coordination call can be as short as 10 to 20 minutes, and it should be open to all the internal and external stakeholders. Calls can be organized and supervised by the Airport Operations Manager. The list of items for this call may vary from one airport to another, but the overall idea is to let the operations lead of each stakeholder give the "heads up" for the next 12 or 24 hours.

In parallel, a status report should be sent daily to a broader range of stakeholders. The status report can serve as minutes of the coordination call. The objective is to make sure that the whole community of operations has access to the information and that records are available for the performance reviews. Templates of the airport status report (.eml and .docx) are available in the "Operate" section of the toolbox.

4.3 ACDM Website

An ACDM website should be created to provide the following:

- Real-time airport operations information (KPIs) at a glance (dashboard).
- Real-time flight information (see section 4.5, Milestone Approach and Departure Metering).
- Specific information on approaching and ongoing adverse conditions.
- Analysis of previous adverse conditions and lessons learned.
- Repertory of ACDM procedures and other documents of reference.

The airport dashboard can also feature a general indicator of the status of operations, as suggested in the daily status report template available in the "Operations" section of the toolbox.

The ACDM website or intranet portal should be accessible to the members of the ACDM initiative only, because it features information shared under the provisions of the ACDM MoU and service-level agreement.

4.4 Integrated Airport Operations Center

Collaboration can be difficult if the stakeholders of the real-time operations are spread around the airport. Great benefits of consolidation or integration of airport operations have been seen at airports in the U.S. and abroad. The concept of an integrated Airport Operations Center—referred to as "APOC" by ICAO—is being adopted by a growing number of hub airports.

The internal stakeholders and some external stakeholders are integrated on the same floor plan and typically share visual displays and indicators on a wide video wall. Such an operations

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center brings all the real-time players of the airport operator under the direct supervision or coordination of the Airport Operations Manager. It increases the coordination with the external stakeholders that can have a workstation at the APOC. It can provide additional facilities to support the collaborative decision-making process, such as a separate meeting room equipped with remote meeting capability and used for the briefings, a decision corner, and other facilities. It calls for defined communication protocols between the APOC and the individual stakeholder operations centers (e.g., airline operations control centers), with both facilities monitoring ATC movements and performance.

4.5 Milestone Approach and Departure Metering

The ATM component of ACDM is not the subject of this project. However, research efforts in the U.S. and achievements of non-U.S. airports already show significant benefits of this component of ACDM that include the reduction of the taxi-in and taxi-out times, an increase of overall punctuality, and noticeable fuel and emission savings.

The main takeaways of the ATM component of ACDM are:

- Segmenting the turnaround process into fundamental elements and defining milestones of the flight turnaround process.
- Optimizing ground operations with a focus on making the aircraft ready for pushback on time and providing notice if more time is needed in revising the EOBT of the flight.
- Switching from a "first called, first served" to a "first scheduled, first served" approach to departure sequencing (FAA Order JO7110.65Y, Paragraph 2-1-4) (FAA 2019).

ACRP Report 137: Guidebook for Advancing Collaborative Decision Making (CDM) at Airports provides more information on these aspects of ACDM and its interface with the existing FAA/Industry CDM process (Vail et al. 2016). FAA Order JO 7110.65Y: *Air Traffic Control* provides procedures and phraseology for use by persons providing ATC services, and it features useful guidance as well (FAA 2019).

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CHAPTER 5

Continuing the ACDM Implementation Effort

5.1 Change Management and Promotion of ACDM

Get Ready for Change

The implementation of ACDM necessarily leads to at least some degree of change in practices, procedures, and even organizations. The success of ACDM implementation relies on a proactive approach to make these changes as smooth as possible for the teams and the operations. Individuals and groups concerned by these changes should be involved with the ACDM projects affecting their work, be prepared to transition to the new processes or technologies, and be trained to operate them.

The proposed comprehensive approach to change management for ACDM implementation is similar to the 6-step process proposed in *ACRP Report 153: Guidebook for IROPS Stakeholder Communication & Coordination* (Cogliandro et al. 2016) (See Figure 10).

1. **Obtain SG buy-in.** In accordance with the process described in the previous chapters, new projects should be proposed by the WG and approved by the SG. The SG then creates TSG to prepare and implement specific projects (changes).

   Submissions to the SG should include at least a description of the project, its goals, a brief assessment of what needs to change and why a list of resources is needed, a budget estimate, and a timeline.
2. **Document existing conditions.** The work of the TSG should always start with an inventory of the existing conditions at the perimeter of the project.
3. **Establish a plan to implement change.** From the existing conditions, the TSG should establish a plan to realize the project and implement the change considering human factors and operational readiness and transition aspects.
4. **Review, revise, and train together.** The TSG proposal should be reviewed (for instance, by the WG) and revised as needed. The parties concerned by the change should train and prepare together for the change. Simulations and drills should be conducted if relevant.
5. **Cooperate during transition.** The stakeholders should cooperate through the transition and activate backup plans if this transition is not completed as planned.
6. **Monitor and plan corrective actions.** Operations should be monitored after the change is implemented for a defined period of time. Corrective actions should be performed if some features of the change appear to be suboptimal or have unforeseen negative impacts.

Involve Your Team with the Projects

Involving the operations team members of the different stakeholders in the ACDM projects they will run will not only allow the projects to benefit from their experience but will also help design and develop an ACDM process that is operationally realistic and efficient. It will contribute

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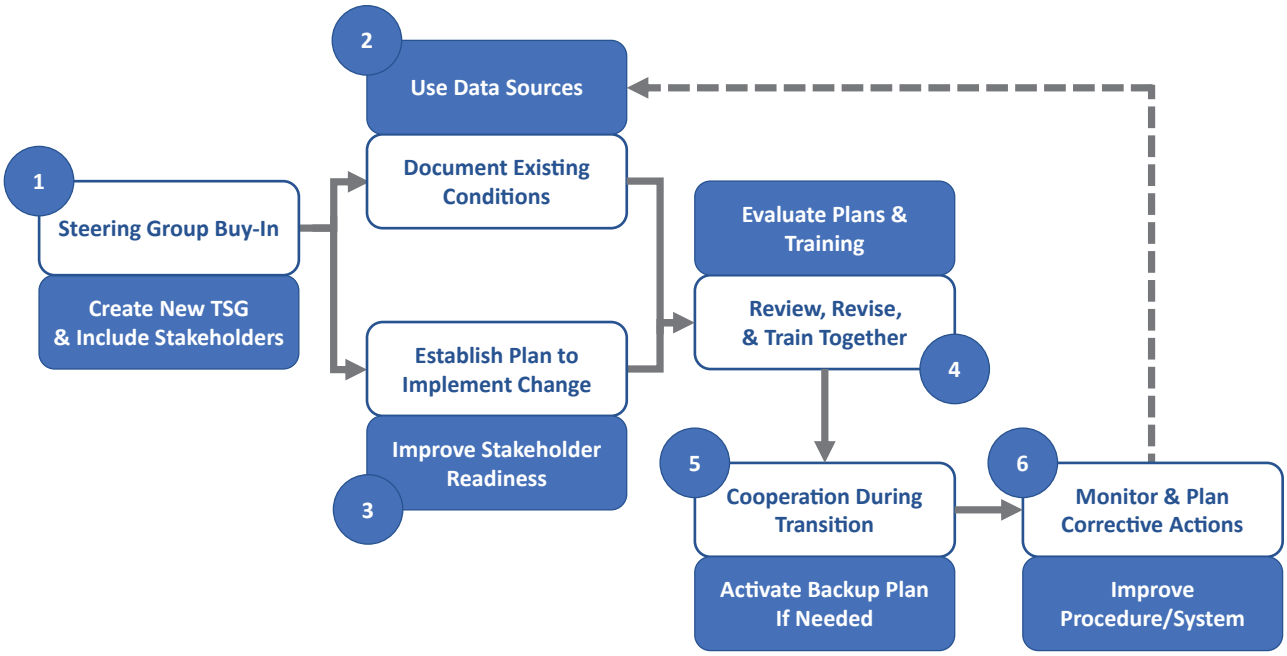


Figure 10. The six-step ACDM change management approach.

to ACDM team building by bringing people together from different entities who will learn about their respective practices, needs, and constraints and will make the participants aware of the coming change. They can then be executors of the change by informing and educating their co-workers.

This involvement with the TSG—in addition to members' regular operational or technical duties—should be considered when estimating the resources needed for each project. Supervisors should be invited to adjust the duty rosters and individual schedules to facilitate the participation of their teams in the ACDM implementation effort and to prevent ACDM from becoming a burden, which could adversely increase the workload.

Consider the Human Side

The introduction of new ACDM procedures and technologies should help the operations in their daily duties and during adverse conditions. According to FAA Order 9550.8: *Human Factor Policy*, human factors entail a multidisciplinary effort to generate and compile information about human capabilities and limitations and apply that information to equipment, systems, facilities, procedures, jobs, environments, training, staffing, and personnel management for safe, comfortable, effective human performance (FAA 1993). The human factor and the potential impact of changes on the individual workload should be considered. Among the stakeholders, the FAA ATCT might be the most familiar with these considerations and can be of tremendous help to the ACDM initiative by taking these aspects into consideration.

Operational Readiness and Transition

Some ACDM projects might require a specific preparation of the stakeholders before their commissioning and sometimes a more comprehensive transition plan. In-depth training sessions, "what if" exercises, and full-scale drills should be conducted for procedures and systems, introducing significant novelties to real-time operations to increase operational readiness.

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Transition plans should include the following:

- A detailed description of the change, mentioning the functions and positions involved,
- An inventory of the different impacts on current practices and organization,
- A risk assessment of these impacts and mitigation when relevant,
- A training plan for the staff members impacted by the change, and
- A backup plan in case the transition is not successful and cannot be complete.

Promoting ACDM to the Airport Operations Community

ACDM promotion is an important component of ACDM implementation. ACDM is a recent concept that is not yet widespread in the United States. In addition, ACDM implementation will introduce several novelties in operations locally. This warrants a specific outreach of the members of the nascent local ACDM community.

ACDM promotion utilizes similar tools and techniques as those used for raising awareness as part of the airport SMS or its security management systems (SeMS). ACDM goodies, newsletters about the updates and latest achievements, regular workshops for getting new ideas from the practitioners, seminars for analyzing recent events, and sharing lessons learned are good ways to keep in touch with the operations community.

The ACDM toolbox itself is important for promoting ACDM. A module of the training program proposed in the "Train" section explains how the toolbox can be shared with the members of the ACDM community as a local library and a repertory of collaborative documents, procedures, templates, and tools.

5.2 Maintaining and Enhancing the Local ACDM

Performance Review Board

A Performance Review Board (PRB) should be established at larger airports. The missions of the PRB can be fulfilled by the SG or an existing body of operations leaders if the creation of a new separate group is not relevant locally.

The goal of this PRB is to review and analyze the airport KPIs over set periods of time (e.g., monthly or quarterly), consider the contribution of the ACDM initiative to their evolution, and provide performance-driven recommendations for enhancing operations overall.

The PRB meetings should focus on the benefits enjoyed by the stakeholders over the time period reviewed, which implies that KPIs measure these benefits either quantitatively or qualitatively. Special sessions can be convened on an ad hoc basis following individual disruptions.

Annual or Biannual ACDM Workshop

The CDM Manager, with the support of the SG, should organize annual or biannual ACDM workshops to reflect on the past period of ACDM operations, gather lessons learned, identify potential enhancements, and strengthen the ACDM spirit (teambuilding) across the stakeholders. The workshops should be times when the ACDM community gathers to discuss how to make the process more efficient and how to resolve operational issues that can make the life of the practitioners easier.

These 1- or 2-day workshops should be interactive and friendly. They can feature thematic roundtables ideally led by moderators from different entities (stakeholders). The participants

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can rotate between the roundtables. The workshops can be preceded by an electronic survey to the local ACDM community for collecting ideas and topics in advance.

The WG should review the output of these workshops and propose to the SG revisions to the ACDM implementation roadmap or individual items of the local ACDM (e.g., plans, procedures, processes) accordingly.

Toward the Next ACDM Roadmap

While the initial ACDM implementation roadmap is being finished, the WG should evaluate and then present the SG with a possible second ACDM implementation roadmap that continues the implementation effort and expands the ACDM initiative. The next ACDM implementation roadmap can keep strategic objectives of the ongoing ACDM implementation roadmap with higher operational objectives or set up new strategic objectives as well. The second ACDM implementation roadmap should not be just an extension of the ongoing ACDM implementation roadmap but should build on the success of the latter in a logic of continuous improvement and cover aspects of operations that are not necessarily the focus of the ongoing initiative.

The design and initiation of the next ACDM implementation roadmap should follow the same process as the initial ACDM implementation roadmap. Other than that, the organizational aspects (Memorandums of Understanding, groups, etc.) are already set. The SGs and WGs should give themselves enough time before the end of the ongoing ACDM implementation roadmap and proactively hold the discussions on the next ACDM implementation roadmap to avoid a lapse of inactivity that could compromise the commitment of the ACDM community.

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CHAPTER 6

Certification Framework

6.1 Purpose

In order to assist airports in their ACDM journey and to ensure consistency throughout the NAS, a certification framework is proposed in the toolbox. This framework is a series of items that are considered critical for the success of a local ACDM process. In other words, a local ACDM process that does not check all the boxes might not be achieving the required level of collaboration and might not have all the features that make an airport ACDM-ready regarding the management of adverse conditions.

Important: The toolbox focuses on managing adverse conditions with ACDM. This certification framework does not consider the air traffic flow management part of ACDM, including technical specifications for real-time data exchange and data quality aspects.

6.2 ACDM Implementation Levels

Three levels of maturity are proposed in the toolbox with regard to the implementation of ACDM (Table 5). These levels are based on the ACDM approach and implementation process proposed in the toolbox.

Note: These levels consider the "Block Upgrades" B0- and B1-ACDM of the Global Air Navigation Plan of the ICAO. Levels 2 and 3 have some features of ICAO's B0- and B1-ACDM, respectively, on the management of adverse conditions.

6.3 Certification Process

The certification of the local ACDM should be conducted either by the WG (self-assessment) or an independent body. Alternatively, the airport industry could agree to trust a professional organization to host and manage the certification process and to provide a third-party, independent audit applying the same consistent judgment nationwide.

The checklist is accessible in the "Operate" section of the toolbox.

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Table 5. Maturity levels of ACDM.

ACDM Maturity Level	Description	Benefits
Level 1 – Enhanced Operations Resilience Through ACDM	Implement collaborative procedures and tools that allow the sharing of airport operations data among the different stakeholders and a decision-making process that allows a concerted response to adverse conditions. This improves the management of adverse conditions and enhances safety, efficiency, and situational awareness.	▪ **Resilience:** Collaborative procedures established through the ACDM planning process and operational readiness achieved through a joint and recurrent training program increase the speed and efficiency of the response to adverse conditions. Stakeholders make decisions collaboratively following the local ACDM operations process in order to achieve a common objective of resilience and efficiency benefiting the community of operations as a whole. ▪ **Efficiency:** The airport is less sensitive to continuity events and emergencies that are proactively managed through the ACDM operations process. ▪ **Capacity:** The ACDM process and the overall enhanced collaboration improve the mutual understanding of individual stakeholder needs and constraints. Coordination mechanisms increase awareness on subsequent issues and enhance the planning and management of resources.
Level 2 – Enhanced Operations Efficiency Through ACDM	Implement collaborative applications that allow the sharing of surface operations data among the different stakeholders at the airport. This improves surface traffic management, reducing delays in both the movement and non-movement areas and enhancing safety, efficiency, and situational awareness.	▪ **Capacity:** The applications result in enhanced use of the existing infrastructure of gates and stands, reduced workload, and better organization of the activities to manage flights. ▪ **Efficiency:** The ATM system is more efficient for all stakeholders; situational awareness on aircraft status is improved, both at the airport and in flight; aircraft fleet predictability and punctuality are enhanced; and operational efficiency is improved. ▪ **Environment:** Reduced taxi times, reduced fuel use and carbon emissions, and lower aircraft engine run times are realized.
Level 3 – Optimized Airport Operations Through ACDM Total Airport Management	Enhance the planning and management of airport operations and allow their full integration in airport operations using performance targets compliant with those of the surrounding airspace. This entails implementing extended collaborative airport operations planning and where needed, an integrated Airport Operations Center (APOC).	▪ **Efficiency:** Through collaborative procedures, comprehensive planning, and proactive response to foreseeable problems, a major reduction in ground and in-flight holding is expected. Actions support the efficient use of resources. Minor increases in resource needs may be expected to support the solutions. ▪ **Predictability:** Through the operational management of performance, reliability and accuracy of the schedule and demand forecast will increase. ▪ **Environment:** Major reductions in ground and in-flight holding are expected, thereby reducing noise and air pollution in the vicinity of airports.

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CHAPTER 7

ACDM Training Programs

7.1 Why Train for ACDM?

ACDM is a new process and requires an adjustment of the state-of-mind of the stakeholders agreeing to work collaboratively. Switching from a pre-ACDM organization to an ACDM one requires a change in the way members of the operations community interact every day. Operations and emergency planners need to learn the ACDM planning process in order to prepare adequate ACDM-compliant plans and procedures. Finally, the members of the ACDM implementation groups should be educated on the different aspects of the program they will run and what they are supposed to deliver.

7.2 ACDM Training Curricula

This section proposes different curricula, and draft training materials are provided. The curricula were designed to fit in with specific roles in the ACDM organization. While these curricula are generic enough to apply to any aviation facility or system, each airport is different, and the person responsible for handling the training program at a specific airport might want to consider adding slides and provisions on local specificities and projects to these materials.

The modules are organized in two different tracks:

1. **Operations (OPS):** Training modules on ACDM operations and planning.
2. **Implementation (IMP):** Training modules on ACDM implementation.

Four curricula are defined for four different roles within the ACDM organization:

- **SG members:** Three mandatory modules are required to understand the basics about ACDM, and five optional modules are recommended for further learning about ACDM operations and implementation.
- **WG members:** The WG members need to be familiar with a broader body of knowledge to manage the ACDM program or individual ACDM projects and to understand ACDM operations as well. It is recommended that the WG members attend all the modules.

  Note: Whatever their full-time role in the ACDM organization, the TSG leads might want to complete their ACDM education by attending all the IMP modules available.
- **ACDM planners:** Members of the operations community in charge of planning for IROPS and emergency situations—and more generally adverse conditions—and those involved with preparing operations plans and procedures should attend six of the OPS modules proposed on ACDM operations and planning. They should also attend the two remaining OPS modules on operating with and maintaining the toolbox.
- **ACDM users:** A similar curriculum focused on real-time operations is proposed to practitioners who might operate with ACDM in real time.

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Table 6. Proposed curricula and modules.

	OPS101	OPS102	OPS103	OPS201	OPS202	OPS203	OPS204	OPS205	IMP101	IMP102	IMP103	IMP104
Steering Group	■								■	■		
Working Group	■	■	■	■	■	■	■	■	■	■	■	■
ACDM Planners	■	■	■	■	■	■						
ACDM Users	■	■		■	■	■						

■ Mandatory Modules (Required) ■ Optional Modules (Recommended)

7.3 Operations Modules

OPS101—Introduction to ACDM

Learning Objectives:

- Understand the main objectives of ACDM.
- Identify the stakeholders of the ACDM process.
- Describe the benefits of ACDM.

OPS102—Operating Together with ACDM

Learning Objectives:

- Identify the different steps of the ACDM operations process.
- Recognize the role of the stakeholders in the ACDM process.
- Operate collaboratively under an ACDM organization.
- Understand the need for operations planning.
- Apply collaborative procedures during real-time operations.

OPS103—ACDM Operations Planning

Learning Objectives:

- Identify the different steps of the ACDM planning process.
- Define the role of the stakeholders in the development of plan and procedures.
- Plan for adverse conditions collaboratively with the stakeholders.
- Collect lessons learned and enhance existing plans and procedures.

OPS201—ACDM Toolbox—Introduction for Practitioners

Learning Objectives:

- Understand the purpose of the toolbox.
- Identify the main components of the toolbox.
- Understand the benefits of the toolbox for planning and operations.

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OPS202—ACDM Toolbox—Templates

Learning Objectives:

- Use the five templates available in the toolbox.
- Understand their benefits to operations.
- Utilize them for enhancing the management of adverse conditions.

OPS203—ACDM Toolbox—Tools

Learning Objectives:

- Use the four tools available in the toolbox.
- Understand their benefits to operations.
- Utilize them for enhancing the management of adverse conditions.

OPS204—ACDM Toolbox—Operate with the Toolbox

Learning Objectives:

- Enhance collaboration between the stakeholders in real-time operations.
- Utilize the toolbox contents to understand the needs of the other stakeholders.
- Utilize the toolbox for planning and operations purposes.

OPS205—ACDM Toolbox—Maintain the Toolbox

Learning Objectives:

- Use the toolbox as a library of local plans and procedures.
- Keep the toolbox up to date.
- Disseminate the toolbox to the stakeholders of airport operations.

7.4 Implementation Modules

IMP101—ACDM for CDM Sponsors and Champions

Learning Objectives:

- Understand the main features of ACDM.
- Promote ACDM within organizations.
- Develop efficient communication channels.

IMP102—ACDM Implementation—Part 1: Initiating ACDM

Learning Objectives:

- Initiate the ACDM implementation project.
- Understand the role of the CDM Champions.
- Develop the ACDM organization.
- Define the basic elements of the ACDM implementation roadmap.

IMP103—ACDM Implementation—Part 2: Developing the Local ACDM

Learning Objectives:

- Develop the local ACDM implementation roadmap.
- Define and monitor the airport KPIs.

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50 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

- Define and implement the local ACDM process.
- Develop collaborative plans and procedures.

IMP104—ACDM Implementation—Part 3: Maintaining the ACDM Initiative

Learning Objectives:

- Develop oversight within the ACDM organization.
- Assess current policies and procedures.
- Review the airport performance and identify potential enhancements.
- Define corrective actions and prioritize them.
- Initiate the development of the next ACDM implementation roadmap.

7.5 Knowledge Management

Keep Track of Knowledge Acquisition

Records of the training modules successfully completed by the members of the ACDM community should be kept and used for ensuring all the participants have received adequate training regarding their role in the organization and the initiative. Certificates of achievement should be provided to participants who successfully complete the module and pass the test.

Knowledge Retention

When customizing the curricula and crafting a local ACDM training program, the ACDM community should think about knowledge retention and how often the participants in the ACDM organization and initiative should retrain on specific modules.

Add Modules to the Curricula

Airports might want to bring training modules from programs outside of the core ACDM basic knowledge under the ACDM curriculum umbrella. Training modules coming from the ACDM planning process on the collaborative plans and procedures to respond to adverse conditions are ideal candidates to add to the ACDM training program. Winter operations training courses involving multiple stakeholders are another example of good candidates. An initiative on joint training and knowledge acquisition and retention across the stakeholders can be a project of the ACDM implementation roadmap by itself.

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APPENDIX A

ACDM Toolbox

Purpose

The toolbox is a "go-to" resource for effectively implementing and using Collaborative Decision Making (CDM) to enhance the management of adverse conditions. The toolbox provides scalable, practice-ready materials that aim to:

- Discuss the benefits of ACDM for mitigating operational disruptions.
- Present best practices, techniques, and tools for engaging stakeholders.
- Provide guidance on how to use data sharing efficiently to improve communication within and between the stakeholders.
- Provide guidance on how to implement the ACDM process at airports—from large-hub airports to smaller facilities, multi-airport systems, statewide aviation systems, and non-traditional airports.
- Provide tools and templates for communication and decision-making purposes.
- Create a consolidated and comprehensive glossary of terminology to help stakeholders understand each other's language and become more familiar with the specific ACDM terminology.

Contents

In the toolbox, practitioners will find contents to:

- **Learn** about ACDM and its benefit for all the stakeholders, including how it can enhance the management of adverse conditions.
- Successfully **Implement** ACDM at an airport, and get the airport operations community ACDM certified.
- **Train** the airport operations community to work and make decisions collaboratively through the ACDM curriculum proposed in the toolbox.
- **Operate** the ACDM process with the Collaborative Decision Sheet providing support for discussion and decision among the stakeholders, with Quick Reference Sheets providing useful information on best practices and case studies, and with various practice-ready templates and spreadsheet-based tools.

The toolbox features a **Glossary**, an **Abbreviations** section, and information about ACRP Project 10-27.

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52 Airport Collaborative Decision Making (ACDM) to Manage Adverse Conditions

How to Use the Toolbox

The toolbox and the toolbox editor are available as a single .ZIP file on the ACRP website. Once unzipped, two folders are available: one with the toolbox and one with the toolbox editor. The toolbox can be accessed by clicking on the file "index.html" inside the toolbox folder. The toolbox is to be opened with a web browser. It is compatible with Google Chrome, Mozilla Firefox, and Apple Safari on Windows, macOS, and Linux.

The toolbox editor is intended to be used by the toolbox manager of an individual ACDM organization to create local versions of the toolbox. (See the "Implement" section in the toolbox and training module OPS205 for more information.)

The toolbox can serve different purposes; some possible utilizations of its contents and resources follow:

1. The sponsors of the ACDM implementation should first read the "Learn" section and then explore the "Implement" section to understand how to design and implement the ACDM process locally.
2. The "Implement" section features detailed guidance on how to organize and manage the ACDM project with the stakeholders.
3. Once the ACDM project is initiated, future users (operating staff) should be trained together following the curriculum proposed in the "Train" section. This curriculum should be customized to individual airports in order to fit in the local context.
4. Once the ACDM process is in place, the CDM Champions should self-certify it following the framework proposed at the end of the "Implement" section.
5. When performing operations planning in real time, the "Operate" section provides insightful case studies and Quick Reference Sheets on a broad range of operational issues where collaboration brings a specific, operations-proven value. The "Operate" section also includes collaborative tools and templates to facilitate the ACDM process with the stakeholders. Training on how to use these items is available in the "Train" section.

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APPENDIX B

Tools and Templates of the Toolbox

The tools and templates listed below are available in the ACDM toolbox. The files can be accessed from the eLibrary and the "Operate" and "Train" sections. Training materials are available in the "Train" section.

Templates for ACDM Implementation

- Memorandum of Understanding (.docx)
- ACDM Implementation Workbook (.pptx)
- Project Progress Report Sheet (.xlsx)
- ACDM Certification Checklist (.xslx)

Templates for Collaborative Operations

- Collaborative Decision Sheet (CDS) (.docx)
- Contact Sheet (.docx)
- Airport Status Report (.eml and .docx)
- Airport Construction Notice (.xlsx)
- Flight Diversion Form (.xlsx)
- Quick Reference Sheet Template (.docx)

Tools for Collaborative Operations

- Flight Operations Planning Simulator (.xslx)
- Aviation System Dashboard (.xslx)
- Aircraft De-icing Capacity Manager (.xslx)
- Regional Diversion Manager (.xslx)

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Abbreviations

ACDM	Airport Collaborative Decision Making
ACRP	Airport Cooperative Research Program
APOC	Airport Operations Center
A-SMS	Airport Safety Management System
ATC	Air Traffic Control
ATCT	Air Traffic Control Tower
ATM	Air Traffic Management
CBP	Customs and Border Protection
CDG	Paris-Charles De Gaulle Airport
CDM	Collaborative Decision Making
DOT	Department of Transportation
FAA	Federal Aviation Administration
ICAO	International Civil Aviation Organization
IMP	Implementation
IOC	Integrated Operations Center
IROPS	Irregular Operations
IT	Information Technology
JFK	John F. Kennedy International Airport
KPI	Key Performance Indicator
MoU	Memorandum of Understanding
NAS	National Airspace System
OPS	Operations
SEADOG	Southeast Airports Disaster Operations Group
SG	Steering Group
SMS	Safety Management Systems
TRACON	Terminal Radar Control
TSA	Transportation Security Administration
TSG	Technical Subgroup
WG	Working Group

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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ISBN 978-0-309-67380-8
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